For *in vivo* or *in vitro* application, the oligonucleotide of the invention is typically capable of inhibiting the expression of the *ATXN3* target nucleic acid in a cell which is expressing the *ATXN3* target nucleic acid. The contiguous sequence of nucleobases of the oligonucleotide of the invention is typically complementary to the *ATXN3* target nucleic acid, as measured across the length of the oligonucleotide, optionally with the exception of one or two mismatches, and optionally excluding nucleotide based linker regions which may link the oligonucleotide to an optional functional group such as a conjugate, or other non-complementary terminal nucleotides (e.g. region D' or D''). The target nucleic acid is a messenger RNA, such as a mature mRNA or a pre-mRNA which encodes mammalian ATXN3 protein, such as human ATXN3, e.g. the human *ATXN3* pre-mRNA sequence, such as that disclosed as SEQ ID NO 1, or *ATXN3* mature mRNA. Further, the target nucleic acid may be a cynomolgus monkey ATXN3 pre-mRNA sequence, such as that disclosed as SEQ ID NO 1, or a cynomolgus monkey ATXN3 mature mRNA. Further, the target nucleic acid may be a mouse ATXN3 pre-mRNA sequence, such as that disclosed as SEQ ID NO 3, or mouse ATXN3 mature mRNA. SEQ ID NOs 1 – 3 are DNA sequences – it will be understood that target RNA sequences have uracil (U) bases in place of the thymidine bases (T).

Table 1

Target Nucleic Acid	**Sequence ID**
ATXN3 *Homo sapiens* pre-mRNA	SEQ ID NO 1
ATXN3 *Macaca fascicularis* pre-mRNA	SEQ ID NO 2
ATXN3 *Mus musculus* mRNA	SEQ ID NO 3

In some embodiments, the oligonucleotide of the invention targets SEQ ID NO 1.

In some embodiments, the oligonucleotide of the invention targets SEQ ID NO 2.

In some embodiments, the oligonucleotide of the invention targets SEQ ID NO 3.

In some embodiments, the oligonucleotide of the invention targets SEQ ID NO 1 and SEQ ID NO 2.

In some embodiments, the oligonucleotide of the invention targets SEQ ID NO 1 and SEQ ID NO 3.

In some embodiments, the oligonucleotide of the invention targets SEQ ID NO 1, SEQ ID NO 2 and SEQ ID NO 3.

Target Sequence

The term "target sequence" as used herein refers to a sequence of nucleotides present in the target nucleic acid which comprises the nucleobase sequence which is complementary to the oligonucleotide of the invention. In some embodiments, the target sequence consists independently comprise 1 - 8 sugar modified nucleosides, and G is a region between 5 and 16 nucleosides which are capable of recruiting RNaseH.

In some embodiments, the sugar modified nucleosides of region F and F' are independently selected from the group consisting of 2'-O-alkyl-RNA, 2'-O-methyl-RNA, 2'-alkoxy-RNA, 2'-O-methoxyethyl-RNA, 2'-amino-DNA, 2'-fluoro-DNA, arabino nucleic acid (ANA), 2'-fluoro-ANA and LNA nucleosides.

In some embodiments, region G comprises 5 – 16 contiguous DNA nucleosides.

In some embodiments, wherein the antisense oligonucleotide is a gapmer oligonucleotide, such as an LNA gapmer oligonucleotide.

In some embodiments, the LNA nucleosides are beta-D-oxy LNA nucleosides.

In some embodiments, the internucleoside linkages between the contiguous nucleotide sequence are phosphorothioate internucleoside linkages.

Preferred sequence motifs and antisense oligonucleotides of the present invention are shown in Table 2 and in the examples.

Table 2: Sequence Motifs and Compounds of Exemplary Compounds of the Invention

SEQ ID NO	motif sequence	start	end	design	CMP ID NO	Oligonucleotide compound
4	aagaaaccaaaccc	743	756	2-10-2	4_1	AAgaaaccaaacCC
5	aaagaaaccaaacc	744	757	2-10-2	5_1	AAagaaaccaaaCC
6	aaaagaaaccaaac	745	758	2-10-2	6_1	AAaagaaaccaaAC
7	caaaagaaaccaaa	746	759	2-10-2	7_1	CAaaagaaaccaAA
8	ccaaaagaaaccaa	747	760	2-10-2	8_1	CCaaaagaaaccAA
9	tccactcctaatac	803	816	2-10-2	9_1	TCcactcctaatAC
10	gtccactcctaata	804	817	2-10-2	10_1	GTccactcctaaTA
11	agtccactcctaat	805	818	2-10-2	11_1	AGtccactcctaAT
12	cagtccactcctaa	806	819	2-10-2	12_1	CAgtccactcctAA
13	ccagtccactccta	807	820	2-10-2	13_1	CCagtccactccTA
14	actctttccaaaca	1012	1025	2-10-2	14_1	ACtctttccaaaCA
15	aactctttccaaac	1013	1026	2-10-2	15_1	AActctttccaaAC
16	caactctttccaaa	1014	1027	2-10-2	16_1	CAactctttccaAA
17	gcaactctttccaa	1015	1028	2-10-2	17_1	GCaactctttccAA
18	agcaactctttcca	1016	1029	2-10-2	18_1	AGcaactctttcCA
19	cagcaactctttcc	1017	1030	2-10-2	19_1	CAgcaactctttCC
20	ccagcaactctttc	1018	1031	2-10-2	20_1	CCagcaactcttTC
21	accagcaactcttt	1019	1032	2-10-2	21_1	ACcagcaactctTT
22	ctcctattaaataa	1040	1053	2-10-2	22_1	CTcctattaaatAA
23	cctcctattaaata	1041	1054	2-10-2	23_1	CCtcctattaaaTA

24	tcctcctattaaat	1042	1055	2-10-2	24_1	TCctcctattaaAT
25	ctcctcctattaaa	1043	1056	2-10-2	25_1	CTcctcctattaAA
26	gctcctcctattaa	1044	1057	2-10-2	26_1	GCtcctcctattAA
27	tgctcctcctatta	1045	1058	2-10-2	27_1	TGctcctcctatTA
28	ttgctcctcctatt	1046	1059	2-10-2	28_1	TTgctcctcctaTT
29	tttgctcctcctat	1047	1060	2-10-2	29_1	TTtgctcctcctAT
30	ctttgctcctccta	1048	1061	2-10-2	30_1	CTttgctcctccTA
31	cctttgctcctcct	1049	1062	2-10-2	31_1	CCtttgctcctcCT
32	ccctttgctcctcc	1050	1063	2-10-2	32_1	CCctttgctcctCC
33	accctttgctcctc	1051	1064	2-10-2	33_1	ACcctttgctccTC
34	aaccctttgctcct	1052	1065	2-10-2	34_1	AAccctttgctcCT
35	aaaccctttgctcc	1053	1066	2-10-2	35_1	AAaccctttgctCC
36	aaaaccctttgctc	1054	1067	2-10-2	36_1	AAaaccctttgcTC
37	aaaaaccctttgct	1055	1068	2-10-2	37_1	AAaaaccctttgCT
38	caaaaaccctttgc	1056	1069	2-10-2	38_1	CAaaaaccctttGC
39	acaaaaaccctttg	1057	1070	2-10-2	39_1	ACaaaaacccttTG
40	aacaaaaaccettt	1058	1071	2-10-2	40_1	AAcaaaaaccctTT
41	aaacaaaaaccctt	1059	1072	2-10-2	41_1	AAacaaaaacccTT
42	aaaacaaaaaccct	1060	1073	2-10-2	42_1	AAaacaaaaaccCT
43	taaaacaaaaaccc	1061	1074	2-10-2	43_1	TAaaacaaaaacCC
44	ataaaacaaaaacc	1062	1075	2-10-2	44_1	ATaaaacaaaaaCC
45	aataaaacaaaaac	1063	1076	2-10-2	45_1	AAtaaaacaaaaAC
46	taataaaacaaaaa	1064	1077	2-10-2	46_1	TAataaaacaaaAA
47	ttaataaaacaaaa	1065	1078	2-10-2	47_1	TTaataaaacaaAA
48	tttaataaaacaaa	1066	1079	2-10-2	48_1	TTtaataaaacaAA
49	atttaataaaacaa	1067	1080	2-10-2	49_1	ATttaataaaacAA
50	ttaaaataaaaatt	1194	1207	2-10-2	50_1	TTaaaataaaaaTT
51	tttaaaataaaaat	1195	1208	2-10-2	51_1	TTtaaaataaaaAT
52	ctttaaaataaaaa	1196	1209	2-10-2	52_1	CTttaaaataaaAA
53	tctttaaaataaaa	1197	1210	2-10-2	53_1	TCtttaaaataaAA
54	atctttaaaataaa	1198	1211	2-10-2	54_1	ATctttaaaataAA
55	catctttaaaataa	1199	1212	2-10-2	55_1	CAtctttaaaatAA
56	ccatctttaaaata	1200	1213	2-10-2	56_1	CCatctttaaaaTA
57	tctaacttaataaa	2886	2899	2-10-2	57_1	TCtaacttaataAA
58	ttctaacttaataa	2887	2900	2-10-2	58_1	TTctaacttaatAA
59	attctaacttaata	2888	2901	2-10-2	59_1	ATtctaacttaaTA
60	cattctaacttaat	2889	2902	2-10-2	60_1	CAttctaacttaAT
61	acattctaacttaa	2890	2903	2-10-2	61_1	ACattctaacttAA
62	tacattctaactta	2891	2904	2-10-2	62_1	TAcattctaactTA
63	ttacattctaactt	2892	2905	2-10-2	63_1	TTacattctaacTT
64	tttacattctaact	2893	2906	2-10-2	64_1	TTtacattctaaCT
65	ttttacattctaac	2894	2907	2-10-2	65_1	TTttacattctaAC

66	tttttacattctaa	2895	2908	2-10-2	66_1	TTtttacattctAA
67	gtttttacattcta	2896	2909	2-10-2	67_1	GTttttacattcTA
68	tgtttttacattct	2897	2910	2-10-2	68_1	TGtttttacattCT
69	ctgtttttacattc	2898	2911	2-10-2	69_1	CTgtttttacatTC
70	ttcaaatatttatt	2969	2982	2-10-2	70_1	TTcaaatatttaTT
71	attcaaatatttat	2970	2983	2-10-2	71_1	ATtcaaatatttAT
72	cattcaaatattta	2971	2984	2-10-2	72_1	CAttcaaatattTA
73	ccattcaaatattt	2972	2985	2-10-2	73_1	CCattcaaatatTT
74	cccattcaaatatt	2973	2986	2-10-2	74_1	CCcattcaaataTT
75	ccccattcaaatat	2974	2987	2-10-2	75_1	CCccattcaaatAT
76	gccccattcaaata	2975	2988	2-10-2	76_1	GCcccattcaaaTA
77	tatacattttttc	3173	3186	2-10-2	77_1	TAtacatttttTC
78	atatacattttttt	3174	3187	2-10-2	78_1	ATatacattttTT
79	tatatacatttttt	3175	3188	2-10-2	79_1	TAtatacattttTT
80	atatatacattttt	3176	3189	2-10-2	80_1	ATatatacatttTT
81	aatatatacatttt	3177	3190	2-10-2	81_1	AAtatatacattTT
82	aaatatatacattt	3178	3191	2-10-2	82_1	AAatatatacatTT
83	caaatatatacatt	3179	3192	2-10-2	83_1	CAaatatatacaTT
84	tcaaatatatacat	3180	3193	2-10-2	84_1	TCaaatatatacAT
85	ttcaaatatataca	3181	3194	2-10-2	85_1	TTcaaatatataCA
86	attcaaatatatac	3182	3195	2-10-2	86_1	ATtcaaatatatAC
87	cattcaaatatata	3183	3196	2-10-2	87_1	CAttcaaatataTA
88	ccattcaaatatat	3184	3197	2-10-2	88_1	CCattcaaatatAT
89	tccattcaaatata	3185	3198	2-10-2	89_1	TCcattcaaataTA
90	atccattcaaatat	3186	3199	2-10-2	90_1	ATccattcaaatAT
91	tatccattcaaata	3187	3200	2-10-2	91_1	TAtccattcaaaTA
92	ttatccattcaaat	3188	3201	2-10-2	92_1	TTatccattcaaAT
93	tttatccattcaaa	3189	3202	2-10-2	93_1	TTtatccattcaAA
94	ctttatccattcaa	3190	3203	2-10-2	94_1	CTttatccattcAA
95	tctttatccattca	3191	3204	2-10-2	95_1	TCtttatccattCA
96	ctctttatccattc	3192	3205	2-10-2	96_1	CTctttatccatTC
97	tctctttatccatt	3193	3206	2-10-2	97_1	TCtctttatccaTT
98	ccatatatatctca	3221	3234	2-10-2	98_1	CCatatatatctCA
99	accatatatatctc	3222	3235	2-10-2	99_1	ACcatatatatcTC
100	caccatatatatct	3223	3236	2-10-2	100_1	CAccatatatatCT
101	gcaccatatatatc	3224	3237	2-10-2	101_1	GCaccatatataTC
102	agcaccatatatat	3225	3238	2-10-2	102_1	AGcaccatatatAT
103	cagcaccatatata	3226	3239	2-10-2	103_1	CAgcaccatataTA
104	acagcaccatatat	3227	3240	2-10-2	104_1	ACagcaccatatAT
105	aacagcaccatata	3228	3241	2-10-2	105_1	AAcagcaccataTA
106	aaaacaaacaacaa	3462	3475	2-10-2	106_1	AAaacaaacaacAA
107	taaaacaaacaaca	3463	3476	2-10-2	107_1	TAaaacaaacaaCA

108	ctaaaacaaacaac	3464	3477	2-10-2	108_1	CTaaaacaaacaAC
109	actaaaacaaacaa	3465	3478	2-10-2	109_1	ACtaaaacaaacAA
110	aactaaaacaaaca	3466	3479	2-10-2	110_1	AActaaaacaaaCA
111	gaactaaaacaaac	3467	3480	2-10-2	111_1	GAactaaaacaaAC
112	agaactaaaacaaa	3468	3481	2-10-2	112_1	AGaactaaaacaAA
113	cagaactaaaacaa	3469	3482	2-10-2	113_1	CAgaactaaaacAA
114	ccagaactaaaaca	3470	3483	2-10-2	114_1	CCagaactaaaaCA
115	accagaactaaaac	3471	3484	2-10-2	115_1	ACcagaactaaaAC
116	atgttattatcccc	3882	3895	2-10-2	116_1	ATgttattatccCC
117	tatgttattatccc	3883	3896	2-10-2	117_1	TAtgttattatcCC
118	ctatgttattatcc	3884	3897	2-10-2	118_1	CTatgttattatCC
119	tctatgttattatc	3885	3898	2-10-2	119_1	TCtatgttattaTC
120	tacactctaactct	3908	3921	2-10-2	120_1	TAcactctaactCT
121	ctacactctaactc	3909	3922	2-10-2	121_1	CTacactctaacTC
122	tctacactctaact	3910	3923	2-10-2	122_1	TCtacactctaaCT
123	ctctacactctaac	3911	3924	2-10-2	123_1	CTctacactctaAC
124	tctctacactctaa	3912	3925	2-10-2	124_1	TCtctacactctAA
125	ttctctacactcta	3913	3926	2-10-2	125_1	TTctctacactcTA
126	cttctctacactct	3914	3927	2-10-2	126_1	CTtctctacactCT
127	ccttctctacactc	3915	3928	2-10-2	127_1	CCttctctacacTC
128	tacaacacaaatca	4102	4115	2-10-2	128_1	TAcaacacaaatCA
129	ctacaacacaaatc	4103	4116	2-10-2	129_1	CTacaacacaaaTC
130	actacaacacaaat	4104	4117	2-10-2	130_1	ACtacaacacaaAT
131	aactacaacacaaa	4105	4118	2-10-2	131_1	AActacaacacaAA
132	taactacaacacaa	4106	4119	2-10-2	132_1	TAactacaacacAA
133	ctaactacaacaca	4107	4120	2-10-2	133_1	CTaactacaacaCA
134	actaactacaacac	4108	4121	2-10-2	134_1	ACtaactacaacAC
135	tactaactacaaca	4109	4122	2-10-2	135_1	TActaactacaaCA
136	ctactaactacaac	4110	4123	2-10-2	136_1	CTactaactacaAC
137	actactaactacaa	4111	4124	2-10-2	137_1	ACtactaactacAA
138	cactactaactaca	4112	4125	2-10-2	138_1	CActactaactaCA
139	acactactaactac	4113	4126	2-10-2	139_1	ACactactaactAC
140	gacactactaacta	4114	4127	2-10-2	140_1	GAcactactaacTA
141	agacactactaact	4115	4128	2-10-2	141_1	AGacactactaaCT
142	tttacccccaacct	4173	4186	2-10-2	142_1	TTtacccccaacCT
143	atttacccccaacc	4174	4187	2-10-2	143_1	ATttacccccaaCC
144	catttacccccaac	4175	4188	2-10-2	144_1	CAtttacccccaAC
145	tcatttacccccaa	4176	4189	2-10-2	145_1	TCatttacccccAA
146	atcatttaccccca	4177	4190	2-10-2	146_1	ATcatttaccccCA
147	aatcatttaccccc	4178	4191	2-10-2	147_1	AAtcatttacccCC
148	aaatcatttacccc	4179	4192	2-10-2	148_1	AAatcatttaccCC
149	caaatcatttaccc	4180	4193	2-10-2	149_1	CAaatcatttacCC

150	ccaaatcatttacc	4181	4194	2-10-2	150_1	CCaaatcatttaCC
151	accaaatcatttac	4182	4195	2-10-2	151_1	ACcaaatcatttAC
152	taccaaatcattta	4183	4196	2-10-2	152_1	TAccaaatcattTA
153	ctaccaaatcattt	4184	4197	2-10-2	153_1	CTaccaaatcatTT
154	gctaccaaatcatt	4185	4198	2-10-2	154_1	GCtaccaaatcaTT
155	tgctaccaaatcat	4186	4199	2-10-2	155_1	TGctaccaaatcAT
156	ctgctaccaaatca	4187	4200	2-10-2	156_1	CTgctaccaaatCA
157	actgctaccaaatc	4188	4201	2-10-2	157_1	ACtgctaccaaaTC
158	aactgctaccaaat	4189	4202	2-10-2	158_1	AActgctaccaaAT
159	aagctttaatcaaa	5102	5115	2-10-2	159_1	AAgctttaatcaAA
160	caagctttaatcaa	5103	5116	2-10-2	160_1	CAagctttaatcAA
161	tcaagctttaatca	5104	5117	2-10-2	161_1	TCaagctttaatCA
162	atcaagctttaatc	5105	5118	2-10-2	162_1	ATcaagctttaaTC
163	catcaagctttaat	5106	5119	2-10-2	163_1	CAtcaagctttaAT
164	tcaaactatcccca	5131	5144	2-10-2	164_1	TCaaactatcccCA
165	ctcaaactatcccc	5132	5145	2-10-2	165_1	CTcaaactatccCC
166	tctcaaactatccc	5133	5146	2-10-2	166_1	TCtcaaactatcCC
167	atctcaaactatcc	5134	5147	2-10-2	167_1	ATctcaaactatCC
168	tatctcaaactatc	5135	5148	2-10-2	168_1	TAtctcaaactaTC
169	ttatctcaaactat	5136	5149	2-10-2	169_1	TTatctcaaactAT
170	cttatctcaaacta	5137	5150	2-10-2	170_1	CTtatctcaaacTA
171	ccttatctcaaact	5138	5151	2-10-2	171_1	CCttatctcaaaCT
172	cccttatctcaaac	5139	5152	2-10-2	172_1	CCcttatctcaaAC
173	gcccttatctcaaa	5140	5153	2-10-2	173_1	GCccttatctcaAA
174	tgcccttatctcaa	5141	5154	2-10-2	174_1	TGcccttatctcAA
175	caaacttcatcaaa	5540	5553	2-10-2	175_1	CAaacttcatcaAA
176	tcaaacttcatcaa	5541	5554	2-10-2	176_1	TCaaacttcatcAA
177	atcaaacttcatca	5542	5555	2-10-2	177_1	ATcaaacttcatCA
178	aatcaaacttcatc	5543	5556	2-10-2	178_1	AAtcaaacttcaTC
179	aaatcaaacttcat	5544	5557	2-10-2	179_1	AAatcaaacttcAT
180	gaaatcaaacttca	5545	5558	2-10-2	180_1	GAaatcaaacttCA
181	tgaaatcaaacttc	5546	5559	2-10-2	181_1	TGaaatcaaactTC
182	ttgaaatcaaactt	5547	5560	2-10-2	182_1	TTgaaatcaaacTT
183	aacacaaatttcct	5693	5706	2-10-2	183_1	AAcacaaatttcCT
184	taacacaaatttcc	5694	5707	2-10-2	184_1	TAacacaaatttCC
185	ctaacacaaatttc	5695	5708	2-10-2	185_1	CTaacacaaattTC
186	gctaacacaaattt	5696	5709	2-10-2	186_1	GCtaacacaaatTT
187	tgctaacacaaatt	5697	5710	2-10-2	187_1	TGctaacacaaaTT
188	ttgctaacacaaat	5698	5711	2-10-2	188_1	TTgctaacacaaAT
189	tttgctaacacaaa	5699	5712	2-10-2	189_1	TTtgctaacacaAA
190	ctttgctaacacaa	5700	5713	2-10-2	190_1	CTttgctaacacAA
191	cctttgctaacaca	5701	5714	2-10-2	191_1	CCtttgctaacaCA

192	taactaataattat	6417	6430	2-10-2	192_1	TAactaataattAT
193	ataactaataatta	6418	6431	2-10-2	193_1	ATaactaataatTA
194	aataactaataatt	6419	6432	2-10-2	194_1	AAtaactaataaTT
195	taataactaataat	6420	6433	2-10-2	195_1	TAataactaataAT
196	ataataactaataa	6421	6434	2-10-2	196_1	ATaataactaatAA
197	aataataactaata	6422	6435	2-10-2	197_1	AAtaataactaaTA
198	caataataactaat	6423	6436	2-10-2	198_1	CAataataactaAT
199	ccaataataactaa	6424	6437	2-10-2	199_1	CCaataataactAA
200	accaataataacta	6425	6438	2-10-2	200_1	ACcaataataacTA
201	aaccaataataact	6426	6439	2-10-2	201_1	AAccaataataaCT
202	taaccaataataac	6427	6440	2-10-2	202_1	TAaccaataataAC
203	ataaccaataataa	6428	6441	2-10-2	203_1	ATaaccaataatAA
204	tataaccaataata	6429	6442	2-10-2	204_1	TAtaaccaataaTA
205	gtataaccaataat	6430	6443	2-10-2	205_1	GTataaccaataAT
206	acatcacacaattt	7415	7428	2-10-2	206_1	ACatcacacaatTT
207	gacatcacacaatt	7416	7429	2-10-2	207_1	GAcatcacacaaTT
208	tgacatcacacaat	7417	7430	2-10-2	208_1	TGacatcacacaAT
209	ctgacatcacacaa	7418	7431	2-10-2	209_1	CTgacatcacacAA
210	tctgacatcacaca	7419	7432	2-10-2	210_1	TCtgacatcacaCA
211	atctgacatcacac	7420	7433	2-10-2	211_1	ATctgacatcacAC
212	ttccttaacccaac	7436	7449	2-10-2	212_1	TTccttaacccaAC
213	attccttaacccaa	7437	7450	2-10-2	213_1	ATtccttaacccAA
214	tattccttaaccca	7438	7451	2-10-2	214_1	TAttccttaaccCA
215	ctattccttaaccc	7439	7452	2-10-2	215_1	CTattccttaacCC
216	tctattccttaacc	7440	7453	2-10-2	216_1	TCtattccttaaCC
217	gtctattccttaac	7441	7454	2-10-2	217_1	GTctattccttaAC
218	catcaaatctcata	8609	8622	2-10-2	218_1	CAtcaaatctcaTA
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220	tgcatcaaatctca	8611	8624	2-10-2	220_1	TGcatcaaatctCA
221	atgcatcaaatctc	8612	8625	2-10-2	221_1	ATgcatcaaatcTC
222	aatgcatcaaatct	8613	8626	2-10-2	222_1	AAtgcatcaaatCT
223	attttaaacaaaca	8637	8650	2-10-2	223_1	ATtttaaacaaaCA
224	tattttaaacaaac	8638	8651	2-10-2	224_1	TAttttaaacaaAC
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228	gaattattttaaac	8642	8655	2-10-2	228_1	GAattattttaaAC
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230	attttacaaatcta	8694	8707	2-10-2	230_1	ATtttacaaatcTA
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232	ttattttacaaatc	8696	8709	2-10-2	232_1	TTattttacaaaTC
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242	aataatttaatcat	9394	9407	2-10-2	242_1	AAtaatttaatcAT
243	aaataatttaatca	9395	9408	2-10-2	243_1	AAataatttaatCA
244	taaataatttaatc	9396	9409	2-10-2	244_1	TAaataatttaaTC
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247	ccctaaataattta	9399	9412	2-10-2	247_1	CCctaaataattTA
248	cccctaaataattt	9400	9413	2-10-2	248_1	CCcctaaataatTT
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263	taaatgccccactc	11007	11020	2-10-2	263_1	TAaatgccccacTC
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273	ctgaaaattcacta	11945	11958	2-10-2	273_1	CTgaaaattcacTA
274	tctgaaaattcact	11946	11959	2-10-2	274_1	TCtgaaaattcaCT
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276	ctactatatacatc	12177	12190	2-10-2	276_1	CTactatatacaTC
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278	gtctactatataca	12179	12192	2-10-2	278_1	GTctactatataCA
279	agtctactatatac	12180	12193	2-10-2	279_1	AGtctactatatAC
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282	actagtctactata	12183	12196	2-10-2	282_1	ACtagtctactaTA
283	aactagtctactat	12184	12197	2-10-2	283_1	AActagtctactAT
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288	tgtatattctaccc	12215	12228	2-10-2	288_1	TGtatattctacCC
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292	aaccacacaattcc	12256	12269	2-10-2	292_1	AAccacacaattCC
293	aaaccacacaattc	12257	12270	2-10-2	293_1	AAaccacacaatTC
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295	gaaaaccacacaat	12259	12272	2-10-2	295_1	GAaaaccacacaAT
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307	tatcactaaatcca	12334	12347	2-10-2	307_1	TAtcactaaatcCA
308	atatcactaaatcc	12335	12348	2-10-2	308_1	ATatcactaaatCC
309	tatatcactaaatc	12336	12349	2-10-2	309_1	TAtatcactaaaTC
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312	agatatatcactaa	12339	12352	2-10-2	312_1	AGatatatcactAA
313	tagatatatcacta	12340	12353	2-10-2	313_1	TAgatatatcacTA
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315	atataaatttctct	12691	12704	2-10-2	315_1	ATataaatttctCT
316	tatataaatttctc	12692	12705	2-10-2	316_1	TAtataaatttcTC
317	atatataaatttct	12693	12706	2-10-2	317_1	ATatataaatttCT

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321	actccattccaaat	12740	12753	2-10-2	321_1	ACtccattccaaAT
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323	ccactccattccaa	12742	12755	2-10-2	323_1	CCactccattccAA
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326	aaaccactccattc	12745	12758	2-10-2	326_1	AAaccactccatTC
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344	cataaaaacccact	13991	14004	2-10-2	344_1	CAtaaaaacccaCT
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411	tgttaatacttttt	16083	16096	2-10-2	411_1	TGttaatactttTT
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452	actatcacatataa	16513	16526	2-10-2	452_1	ACtatcacatatAA
453	cactatcacatata	16514	16527	2-10-2	453_1	CActatcacataTA
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457	tcagtccaacataa	16837	16850	2-10-2	457_1	TCagtccaacatAA
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468	taaacattaaaccc	16929	16942	2-10-2	468_1	TAaacattaaacCC
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685	aaataatccaaact	27773	27786	2-10-2	685_1	AAataatccaaaCT
686	aaaataatccaaac	27774	27787	2-10-2	686_1	AAaataatccaaAC
687	caaaataatccaaa	27775	27788	2-10-2	687_1	CAaaataatccaAA
688	acaaaataatccaa	27776	27789	2-10-2	688_1	ACaaaataatccAA
689	tacaaaataatcca	27777	27790	2-10-2	689_1	TAcaaaataatcCA
690	ttacaaaataatcc	27778	27791	2-10-2	690_1	TTacaaaataatCC
691	gttacaaaataatc	27779	27792	2-10-2	691_1	GTtacaaaataaTC
692	tgttacaaaataat	27780	27793	2-10-2	692_1	TGttacaaaataAT
693	ttttacattaacta	27935	27948	2-10-2	693_1	TTttacattaacTA
694	tttttacattaact	27936	27949	2-10-2	694_1	TTtttacattaaCT
695	ttttttacattaac	27937	27950	2-10-2	695_1	TTttttacattaAC

696	attttttacattaa	27938	27951	2-10-2	696_1	ATttttacattAA
697	tattttttacatta	27939	27952	2-10-2	697_1	TAtttttacatTA
698	ttattttttacatt	27940	27953	2-10-2	698_1	TTatttttacaTT
699	aaatactaacatca	29299	29312	2-10-2	699_1	AAatactaacatCA
700	aaaatactaacatc	29300	29313	2-10-2	700_1	AAaatactaacaTC
701	caaaatactaacat	29301	29314	2-10-2	701_1	CAaaatactaacAT
702	ccaaaatactaaca	29302	29315	2-10-2	702_1	CCaaaatactaaCA
703	gccaaaatactaac	29303	29316	2-10-2	703_1	GCcaaaatactaAC
704	tgccaaaatactaa	29304	29317	2-10-2	704_1	TGccaaaatactAA
705	tccattcattttat	29415	29428	2-10-2	705_1	TCcattcattttAT
706	atccattcatttta	29416	29429	2-10-2	706_1	ATccattcatttTA
707	catccattcatttt	29417	29430	2-10-2	707_1	CAtccattcattTT
708	acatccattcattt	29418	29431	2-10-2	708_1	ACatccattcatTT
709	cacatccattcatt	29419	29432	2-10-2	709_1	CAcatccattcaTT
710	ccacatccattcat	29420	29433	2-10-2	710_1	CCacatccattcAT
711	gccacatccattca	29421	29434	2-10-2	711_1	GCcacatccattCA
712	tgccacatccattc	29422	29435	2-10-2	712_1	TGccacatccatTC
713	atgccacatccatt	29423	29436	2-10-2	713_1	ATgccacatccaTT
714	tatgccacatccat	29424	29437	2-10-2	714_1	TAtgccacatccAT
715	ttatgccacatcca	29425	29438	2-10-2	715_1	TTatgccacatcCA
716	attatgccacatcc	29426	29439	2-10-2	716_1	ATtatgccacatCC
717	tcttaactcttctc	30753	30766	2-10-2	717_1	TCttaactcttcTC
718	ttcttaactcttct	30754	30767	2-10-2	718_1	TTcttaactcttCT
719	gttcttaactcttc	30755	30768	2-10-2	719_1	GTtcttaactctTC
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721	tagttcttaactct	30757	30770	2-10-2	721_1	TAgttcttaactCT
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723	tcaaatactcaaaa	31030	31043	2-10-2	723_1	TCaaatactcaaAA
724	ttcaaatactcaaa	31031	31044	2-10-2	724_1	TTcaaatactcaAA
725	cttcaaatactcaa	31032	31045	2-10-2	725_1	CTtcaaatactcAA
726	gcttcaaatactca	31033	31046	2-10-2	726_1	GCttcaaatactCA
727	agcttcaaatactc	31034	31047	2-10-2	727_1	AGcttcaaatacTC
728	aagcttcaaatact	31035	31048	2-10-2	728_1	AAgcttcaaataCT
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730	tcctcattacccat	32060	32073	2-10-2	730_1	TCctcattacccAT
731	atcctcattaccca	32061	32074	2-10-2	731_1	ATcctcattaccCA
732	tatcctcattaccc	32062	32075	2-10-2	732_1	TAtcctcattacCC
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734	aatatcctcattac	32064	32077	2-10-2	734_1	AAtatcctcattAC
735	taatatcctcatta	32065	32078	2-10-2	735_1	TAatatcctcatTA
736	ttaatatcctcatt	32066	32079	2-10-2	736_1	TTaatatcctcaTT
737	tttaatatcctcat	32067	32080	2-10-2	737_1	TTtaatatcctcAT

738	atttaatatcctca	32068	32081	2-10-2	738_1	ATttaatatcctCA
739	aatttaatatcctc	32069	32082	2-10-2	739_1	AAtttaatatccTC
740	aaatttaatatcct	32070	32083	2-10-2	740_1	AAatttaatatcCT
741	taaatttaatatcc	32071	32084	2-10-2	741_1	TAaatttaatatCC
742	ttaaatttaatatc	32072	32085	2-10-2	742_1	TTaaatttaataTC
743	cttaaatttaatat	32073	32086	2-10-2	743_1	CTtaaatttaatAT
744	tcttaaatttaata	32074	32087	2-10-2	744_1	TCttaaatttaaTA
745	ttcttaaatttaat	32075	32088	2-10-2	745_1	TTcttaaatttaAT
746	gttcttaaatttaa	32076	32089	2-10-2	746_1	GTtcttaaatttAA
747	ttattctactttta	33431	33444	2-10-2	747_1	TTattctactttTA
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749	ctttattctacttt	33433	33446	2-10-2	749_1	CTttattctactTT
750	cctttattctactt	33434	33447	2-10-2	750_1	CCtttattctacTT
751	gcctttattctact	33435	33448	2-10-2	751_1	GCctttattctaCT
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753	caacaattattaat	33798	33811	2-10-2	753_1	CAacaattattaAT
754	gcaacaattattaa	33799	33812	2-10-2	754_1	GCaacaattattAA
755	agcaacaattatta	33800	33813	2-10-2	755_1	AGcaacaattatTA
756	cagcaacaattatt	33801	33814	2-10-2	756_1	CAgcaacaattaTT
757	ccagcaacaattat	33802	33815	2-10-2	757_1	CCagcaacaattAT
758	accagcaacaatta	33803	33816	2-10-2	758_1	ACcagcaacaatTA
759	aaaccaaaacttac	33963	33976	2-10-2	759_1	AAaccaaaacttAC
760	aaaaccaaaactta	33964	33977	2-10-2	760_1	AAaaccaaaactTA
761	aaaaaccaaaactt	33965	33978	2-10-2	761_1	AAaaaccaaaacTT
762	caaaaaccaaaact	33966	33979	2-10-2	762_1	CAaaaaccaaaaCT
763	ccaaaaaccaaaac	33967	33980	2-10-2	763_1	CCaaaaaccaaaAC
764	accaaaaaccaaaa	33968	33981	2-10-2	764_1	ACcaaaaaccaaAA
765	aaccaaaaaccaaa	33969	33982	2-10-2	765_1	AAccaaaaaccaAA
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768	aatctaaaacactt	34051	34064	2-10-2	768_1	AAtctaaaacacTT
769	aaatctaaaacact	34052	34065	2-10-2	769_1	AAatctaaaacaCT
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771	ccaaatctaaaaca	34054	34067	2-10-2	771_1	CCaaatctaaaaCA
772	cccaaatctaaaac	34055	34068	2-10-2	772_1	CCcaaatctaaaAC
773	ccccaaatctaaaa	34056	34069	2-10-2	773_1	CCccaaatctaaAA
774	accccaaatctaaa	34057	34070	2-10-2	774_1	ACcccaaatctaAA
775	aaccccaaatctaa	34058	34071	2-10-2	775_1	AAccccaaatctAA
776	aaaccccaaatcta	34059	34072	2-10-2	776_1	AAaccccaaatcTA
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779	atattcacaaatcc	34077	34090	2-10-2	779_1	ATattcacaaatCC

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781	aaatattcacaaat	34079	34092	2-10-2	781_1	AAatattcacaaAT
782	caaatattcacaaa	34080	34093	2-10-2	782_1	CAaatattcacaAA
783	gcaaatattcacaa	34081	34094	2-10-2	783_1	GCaaatattcacAA
784	aacacacattatca	34537	34550	2-10-2	784_1	AAcacacattatCA
785	taacacacattatc	34538	34551	2-10-2	785_1	TAacacacattaTC
786	ttaacacacattat	34539	34552	2-10-2	786_1	TTaacacacattAT
787	tttaacacacatta	34540	34553	2-10-2	787_1	TTtaacacacatTA
788	atttaacacacatt	34541	34554	2-10-2	788_1	ATttaacacacaTT
789	tatttaacacacat	34542	34555	2-10-2	789_1	TAtttaacacacAT
790	ctatttaacacaca	34543	34556	2-10-2	790_1	CTatttaacacaCA
791	actatttaacacac	34544	34557	2-10-2	791_1	ACtatttaacacAC
792	tactatttaacaca	34545	34558	2-10-2	792_1	TActatttaacaCA
793	ctactatttaacac	34546	34559	2-10-2	793_1	CTactatttaacAC
794	actactatttaaca	34547	34560	2-10-2	794_1	ACtactatttaaCA
795	aactactatttaac	34548	34561	2-10-2	795_1	AActactatttaAC
796	aaactactatttaa	34549	34562	2-10-2	796_1	AAactactatttAA
797	aaaactactattta	34550	34563	2-10-2	797_1	AAaactactattTA
798	gaaaactactattt	34551	34564	2-10-2	798_1	GAaaactactatTT
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803	acaaaataacctat	35312	35325	2-10-2	803_1	ACaaaataacctAT
804	cacaaaataaccta	35313	35326	2-10-2	804_1	CAcaaaataaccTA
805	tcacaaaataacct	35314	35327	2-10-2	805_1	TCacaaaataacCT
806	atcacaaaataacc	35315	35328	2-10-2	806_1	ATcacaaaataaCC
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808	tcatcacaaaataa	35317	35330	2-10-2	808_1	TCatcacaaaatAA
809	ttcatcacaaaata	35318	35331	2-10-2	809_1	TTcatcacaaaaTA
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812	attttcatcacaaa	35321	35334	2-10-2	812_1	ATtttcatcacaAA
813	tattttcatcacaa	35322	35335	2-10-2	813_1	TAttttcatcacAA
814	gtattttcatcaca	35323	35336	2-10-2	814_1	GTattttcatcaCA
815	atttaaatttatca	35354	35367	2-10-2	815_1	ATttaaatttatCA
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817	aaatttaaatttat	35356	35369	2-10-2	817_1	AAatttaaatttAT
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819	taaaatttaaattt	35358	35371	2-10-2	819_1	TAaaatttaaatTT
820	ataaaatttaaatt	35359	35372	2-10-2	820_1	ATaaaatttaaaTT
821	cataaaatttaaat	35360	35373	2-10-2	821_1	CAtaaaatttaaAT

822	acataaaatttaaa	35361	35374	2-10-2	822_1	ACataaaatttaAA
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824	cctactaatattca	36333	36346	2-10-2	824_1	CCtactaatattCA
825	acctactaatattc	36334	36347	2-10-2	825_1	ACctactaatatTC
826	cacctactaatatt	36335	36348	2-10-2	826_1	CAcctactaataTT
827	tcacctactaatat	36336	36349	2-10-2	827_1	TCacctactaatAT
828	ttcacctactaata	36337	36350	2-10-2	828_1	TTcacctactaaTA
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833	tatttttcacctac	36342	36355	2-10-2	833_1	TAtttttcacctAC
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859	ttacattcccatta	36791	36804	2-10-2	859_1	TTacattcccatTA
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862	cttttacattccca	36794	36807	2-10-2	862_1	CTtttacattccCA
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904	tacttcttttatca	40157	40170	2-10-2	904_1	TActtcttttatCA
905	ttacttcttttatc	40158	40171	2-10-2	905_1	TTacttcttttaTC

906	attacttcttttat	40159	40172	2-10-2	906_1	ATtacttcttttAT
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908	aaattacttctttt	40161	40174	2-10-2	908_1	AAattacttcttTT
909	aaaattacttcttt	40162	40175	2-10-2	909_1	AAaattacttctTT
910	caaaattacttctt	40163	40176	2-10-2	910_1	CAaaattacttcTT
911	ccaaaattacttct	40164	40177	2-10-2	911_1	CCaaaattacttCT
912	tccaaaattacttc	40165	40178	2-10-2	912_1	TCcaaaattactTC
913	ttccaaaattactt	40166	40179	2-10-2	913_1	TTccaaaattacTT
914	gttccaaaattact	40167	40180	2-10-2	914_1	GTtccaaaattaCT
915	tgttccaaaattac	40168	40181	2-10-2	915_1	TGttccaaaattAC
916	atgttccaaaatta	40169	40182	2-10-2	916_1	ATgttccaaaatTA
917	ttactctttttatt	40201	40214	2-10-2	917_1	TTactctttttaTT
918	tttactctttttat	40202	40215	2-10-2	918_1	TTtactcttttTAT
919	ttttactcttttta	40203	40216	2-10-2	919_1	TTttactcttttTA
920	attttactcttttt	40204	40217	2-10-2	920_1	ATtttactctttTT
921	tattttactctttt	40205	40218	2-10-2	921_1	TAttttactcttTT
922	atattttactcttt	40206	40219	2-10-2	922_1	ATattttactctTT
923	catattttactctt	40207	40220	2-10-2	923_1	CAtattttactcTT
924	ccatattttactct	40208	40221	2-10-2	924_1	CCatattttactCT
925	cccatattttactc	40209	40222	2-10-2	925_1	CCcatattttacTC
926	acccatattttact	40210	40223	2-10-2	926_1	ACccatattttaCT
927	tacccatattttac	40211	40224	2-10-2	927_1	TAcccatattttAC
928	ttacccatatttta	40212	40225	2-10-2	928_1	TTacccatatttTA
929	tttacccatatttt	40213	40226	2-10-2	929_1	TTtacccatattTT
930	gtttacccatattt	40214	40227	2-10-2	930_1	GTttacccatatTT
931	tgtttacccatatt	40215	40228	2-10-2	931_1	TGtttacccataTT
932	gttacctcccttta	40368	40381	2-10-2	932_1	GTtacctcccttTA
933	ggttacctcccttt	40369	40382	2-10-2	933_1	GGttacctccctTT
934	aggttacctccctt	40370	40383	2-10-2	934_1	AGgttacctcccTT
935	caaactaaaaccta	41659	41672	2-10-2	935_1	CAaactaaaaccTA
936	tcaaactaaaacct	41660	41673	2-10-2	936_1	TCaaactaaaacCT
937	atcaaactaaaacc	41661	41674	2-10-2	937_1	ATcaaactaaaaCC
938	gatcaaactaaaac	41662	41675	2-10-2	938_1	GAtcaaactaaaAC
939	agatcaaactaaaa	41663	41676	2-10-2	939_1	AGatcaaactaaAA
940	aagatcaaactaaa	41664	41677	2-10-2	940_1	AAgatcaaactaAA
941	ccaatttcacccaa	41699	41712	2-10-2	941_1	CCaatttcacccAA
942	cccaatttcaccca	41700	41713	2-10-2	942_1	CCcaatttcaccCA
943	gcccaatttcaccc	41701	41714	2-10-2	943_1	GCccaatttcacCC
944	tgcccaatttcacc	41702	41715	2-10-2	944_1	TGcccaatttcaCC
945	ttgcccaatttcac	41703	41716	2-10-2	945_1	TTgcccaatttcAC
946	caactttctatttt	41777	41790	2-10-2	946_1	CAactttctattTT
947	ccaactttctattt	41778	41791	2-10-2	947_1	CCaactttctatTT

948	cccaactttctatt	41779	41792	2-10-2	948_1	CCcaactttctaTT
949	acccaactttctat	41780	41793	2-10-2	949_1	ACccaactttctAT
950	aacccaactttcta	41781	41794	2-10-2	950_1	AAcccaactttcTA
951	aaacccaactttct	41782	41795	2-10-2	951_1	AAacccaactttCT
952	aaaacccaactttc	41783	41796	2-10-2	952_1	AAaacccaacttTC
953	aaaaacccaacttt	41784	41797	2-10-2	953_1	AAaaacccaactTT
954	caaaaacccaactt	41785	41798	2-10-2	954_1	CAaaaacccaacTT
955	acaaaaacccaact	41786	41799	2-10-2	955_1	ACaaaaacccaaCT
956	ctttaaaatttcca	42170	42183	2-10-2	956_1	CTttaaaatttcCA
957	tctttaaaatttcc	42171	42184	2-10-2	957_1	TCtttaaaatttCC
958	ttctttaaaatttc	42172	42185	2-10-2	958_1	TTctttaaaattTC
959	tttctttaaaattt	42173	42186	2-10-2	959_1	TTtctttaaaatTT
960	atttctttaaaatt	42174	42187	2-10-2	960_1	ATttctttaaaaTT
961	catttctttaaaat	42175	42188	2-10-2	961_1	CAtttctttaaaAT
962	acatttctttaaaa	42176	42189	2-10-2	962_1	ACatttctttaaAA
963	cacatttctttaaa	42177	42190	2-10-2	963_1	CAcatttctttaAA
964	ccacatttctttaa	42178	42191	2-10-2	964_1	CCacatttctttAA
965	accacatttcttta	42179	42192	2-10-2	965_1	ACcacatttcttTA
966	aaccacatttcttt	42180	42193	2-10-2	966_1	AAccacatttctTT
967	aaaccacatttctt	42181	42194	2-10-2	967_1	AAaccacatttcTT
968	aaaaccacatttct	42182	42195	2-10-2	968_1	AAaaccacatttCT
969	caaaaccacatttc	42183	42196	2-10-2	969_1	CAaaaccacattTC
970	ttcttctcttttca	43831	43844	2-10-2	970_1	TTcttctcttttCA
971	tttcttctcttttc	43832	43845	2-10-2	971_1	TTtcttctctttTC
972	ttttcttctctttt	43833	43846	2-10-2	972_1	TTttcttctcttTT
973	tttttcttctcttt	43834	43847	2-10-2	973_1	TTtttcttctctTT
974	ttttttcttctctt	43835	43848	2-10-2	974_1	TTttttcttctcTT
975	atttttttcttctct	43836	43849	2-10-2	975_1	ATttttttcttctCT
976	tatttttcttctc	43837	43850	2-10-2	976_1	TAtttttcttcTC
977	aacttaatattaaa	45488	45501	2-10-2	977_1	AActtaatattaAA
978	caacttaatattaa	45489	45502	2-10-2	978_1	CAacttaatattAA
979	tcaacttaatatta	45490	45503	2-10-2	979_1	TCaacttaatatTA
980	ttcaacttaatatt	45491	45504	2-10-2	980_1	TTcaacttaataTT
981	attcaacttaatat	45492	45505	2-10-2	981_1	ATtcaacttaatAT
982	tattcaacttaata	45493	45506	2-10-2	982_1	TAttcaacttaaTA
983	ttattcaacttaat	45494	45507	2-10-2	983_1	TTattcaacttaAT
984	tttattcaacttaa	45495	45508	2-10-2	984_1	TTtattcaacttAA
985	caaattaaaaaaca	47397	47410	2-10-2	985_1	CAaattaaaaaaCA
986	tcaaattaaaaaac	47398	47411	2-10-2	986_1	TCaaattaaaaaAC
987	ttcaaattaaaaaa	47399	47412	2-10-2	987_1	TTcaaattaaaaAA
988	cttcaaattaaaaa	47400	47413	2-10-2	988_1	CTtcaaattaaaAA
989	tcttcaaattaaaa	47401	47414	2-10-2	989_1	TCttcaaattaaAA

990	ttcttcaaattaaa	47402	47415	2-10-2	990_1	TTcttcaaattaAA
991	tttcttcaaattaa	47403	47416	2-10-2	991_1	TTtcttcaaattAA
992	aacacaaattcaaa	48077	48090	2-10-2	992_1	AAcacaaattcaAA
993	aaacacaaattcaa	48078	48091	2-10-2	993_1	AAacacaaattcAA
994	taaacacaaattca	48079	48092	2-10-2	994_1	TAaacacaaattCA
995	ataaacacaaattc	48080	48093	2-10-2	995_1	ATaaacacaaatTC
996	aataaacacaaatt	48081	48094	2-10-2	996_1	AAtaaacacaaaTT
997	caataaacacaaat	48082	48095	2-10-2	997_1	CAataaacacaaAT
998	acaataaacacaaa	48083	48096	2-10-2	998_1	ACaataaacacaAA
999	aacaataaacacaa	48084	48097	2-10-2	999_1	AAcaataaacacAA
1000	taacaataaacaca	48085	48098	2-10-2	1000_1	TAacaataaacaCA
1001	ttaacaataaacac	48086	48099	2-10-2	1001_1	TTaacaataaacAC
1002	attaacaataaaca	48087	48100	2-10-2	1002_1	ATtaacaataaaCA
1003	aattaacaataaac	48088	48101	2-10-2	1003_1	AAttaacaataaAC
1004	gaattaacaataaa	48089	48102	2-10-2	1004_1	GAattaacaataAA
1005	tgaattaacaataa	48090	48103	2-10-2	1005_1	TGaattaacaatAA
1006	atattcctcaatca	48905	48918	2-10-2	1006_1	ATattcctcaatCA
1007	tatattcctcaatc	48906	48919	2-10-2	1007_1	TAtattcctcaaTC
1008	atatattcctcaat	48907	48920	2-10-2	1008_1	ATatattcctcaAT
1009	aatatattcctcaa	48908	48921	2-10-2	1009_1	AAtatattcctcAA
1010	caatatattcctca	48909	48922	2-10-2	1010_1	CAatatattcctCA
1011	acaatatattcctc	48910	48923	2-10-2	1011_1	ACaatatattccTC
1012	gacaatatattcct	48911	48924	2-10-2	1012_1	GAcaatatattcCT
1013	caatcctaattaaa	48960	48973	2-10-2	1013_1	CAatcctaattaAA
1014	ccaatcctaattaa	48961	48974	2-10-2	1014_1	CCaatcctaattAA
1015	cccaatcctaatta	48962	48975	2-10-2	1015_1	CCcaatcctaatTA
1016	gcccaatcctaatt	48963	48976	2-10-2	1016_1	GCccaatcctaaTT
1017	tgcccaatcctaat	48964	48977	2-10-2	1017_1	TGcccaatcctaAT
1018	accctacaaatact	50093	50106	2-10-2	1018_1	ACcctacaaataCT
1019	aaccctacaaatac	50094	50107	2-10-2	1019_1	AAccctacaaatAC
1020	aaaccctacaaata	50095	50108	2-10-2	1020_1	AAaccctacaaaTA
1021	aaaaccctacaaat	50096	50109	2-10-2	1021_1	AAaaccctacaaAT
1022	aaaaaccctacaaa	50097	50110	2-10-2	1022_1	AAaaaccctacaAA
1023	aaaaaaccctacaa	50098	50111	2-10-2	1023_1	AAaaaaccctacAA
1024	aaaaaaaccctaca	50099	50112	2-10-2	1024_1	AAaaaaaccctaCA
1025	tatacactattaat	51008	51021	2-10-2	1025_1	TAtacactattaAT
1026	ttatacactattaa	51009	51022	2-10-2	1026_1	TTatacactattAA
1027	attatacactatta	51010	51023	2-10-2	1027_1	ATtatacactatTA
1028	aattatacactatt	51011	51024	2-10-2	1028_1	AAttatacactaTT
1029	gaattatacactat	51012	51025	2-10-2	1029_1	GAattatacactAT
1030	gtaacaattataca	51866	51879	2-10-2	1030_1	GTaacaattataCA
1031	tgtaacaattatac	51867	51880	2-10-2	1031_1	TGtaacaattatAC

1032	ctgtaacaattata	51868	51881	2-10-2	1032_1	CTgtaacaattaTA
1033	cctgtaacaattat	51869	51882	2-10-2	1033_1	CCtgtaacaattAT
1034	tcctgtaacaatta	51870	51883	2-10-2	1034_1	TCctgtaacaatTA
1035	ataaaaaccacctt	53263	53276	2-10-2	1035_1	ATaaaaaccaccTT
1036	aataaaaaccacct	53264	53277	2-10-2	1036_1	AAtaaaaaccacCT
1037	gaataaaaaccacc	53265	53278	2-10-2	1037_1	GAataaaaaccaCC
1038	agaataaaaaccac	53266	53279	2-10-2	1038_1	AGaataaaaaccAC
1039	cagaataaaaacca	53267	53280	2-10-2	1039_1	CAgaataaaaacCA
1040	ccagaataaaaacc	53268	53281	2-10-2	1040_1	CCagaataaaaaCC
1041	cccagaataaaaac	53269	53282	2-10-2	1041_1	CCcagaataaaaAC
1042	acccagaataaaaa	53270	53283	2-10-2	1042_1	ACccagaataaaAA
1043	tttcttactcccct	53699	53712	2-10-2	1043_1	TTtcttactcccCT
1044	ctttcttactcccc	53700	53713	2-10-2	1044_1	CTttcttactccCC
1045	actttcttactccc	53701	53714	2-10-2	1045_1	ACtttcttactcCC
1046	cactttcttactcc	53702	53715	2-10-2	1046_1	CActttcttactCC
1047	ccactttcttactc	53703	53716	2-10-2	1047_1	CCactttcttacTC
1048	cctttaccactttt	53948	53961	2-10-2	1048_1	CCtttaccacttTT
1049	ccctttaccacttt	53949	53962	2-10-2	1049_1	CCctttaccactTT
1050	tccctttaccactt	53950	53963	2-10-2	1050_1	TCcctttaccacTT
1051	atccctttaccact	53951	53964	2-10-2	1051_1	ATccctttaccaCT
1052	catccctttaccac	53952	53965	2-10-2	1052_1	CAtccctttaccAC
1053	ctacatctaacccc	54550	54563	2-10-2	1053_1	CTacatctaaccCC
1054	tctacatctaaccc	54551	54564	2-10-2	1054_1	TCtacatctaacCC
1055	gtctacatctaacc	54552	54565	2-10-2	1055_1	GTctacatctaaCC
1056	agtctacatctaac	54553	54566	2-10-2	1056_1	AGtctacatctaAC
1057	cagtctacatctaa	54554	54567	2-10-2	1057_1	CAgtctacatctAA
1058	tcagtctacatcta	54555	54568	2-10-2	1058_1	TCagtctacatcTA
1059	ttcagtctacatct	54556	54569	2-10-2	1059_1	TTcagtctacatCT
1060	taaccacacctcct	54573	54586	2-10-2	1060_1	TAaccacacctcCT
1061	ttaaccacacctcc	54574	54587	2-10-2	1061_1	TTaaccacacctCC
1062	tttaaccacacctc	54575	54588	2-10-2	1062_1	TTtaaccacaccTC
1063	ttttaaccacacct	54576	54589	2-10-2	1063_1	TTttaaccacacCT
1064	gttttaaccacacc	54577	54590	2-10-2	1064_1	GTtttaaccacaCC
1065	agttttaaccacac	54578	54591	2-10-2	1065_1	AGttttaaccacAC
1066	caacaaaacatcaa	55228	55241	2-10-2	1066_1	CAacaaaacatcAA
1067	tcaacaaaacatca	55229	55242	2-10-2	1067_1	TCaacaaaacatCA
1068	ttcaacaaaacatc	55230	55243	2-10-2	1068_1	TTcaacaaaacaTC
1069	tttcaacaaaacat	55231	55244	2-10-2	1069_1	TTtcaacaaaacAT
1070	ttttcaacaaaaca	55232	55245	2-10-2	1070_1	TTttcaacaaaaCA
1071	gttttcaacaaaac	55233	55246	2-10-2	1071_1	GTtttcaacaaaAC
1072	tgttttcaacaaaa	55234	55247	2-10-2	1072_1	TGttttcaacaaAA
1073	ttctaaaacttacc	55269	55282	2-10-2	1073_1	TTctaaaacttaCC

1074	tttctaaaacttac	55270	55283	2-10-2	1074_1	TTtctaaaacttAC
1075	ctttctaaaactta	55271	55284	2-10-2	1075_1	CTttctaaaactTA
1076	tctttctaaaactt	55272	55285	2-10-2	1076_1	TCtttctaaaacTT
1077	atctttctaaaact	55273	55286	2-10-2	1077_1	ATctttctaaaaCT
1078	aatctttctaaaac	55274	55287	2-10-2	1078_1	AAtctttctaaaAC
1079	gaatctttctaaaa	55275	55288	2-10-2	1079_1	GAatctttctaaAA
1080	agaatctttctaaa	55276	55289	2-10-2	1080_1	AGaatctttctaAA
1081	cagaatctttctaa	55277	55290	2-10-2	1081_1	CAgaatctttctAA
1082	cctttatttccctt	55494	55507	2-10-2	1082_1	CCtttatttcccTT
1083	ccctttatttccct	55495	55508	2-10-2	1083_1	CCctttatttccCT
1084	tccctttatttccc	55496	55509	2-10-2	1084_1	TCcctttatttcCC
1085	ttccctttatttcc	55497	55510	2-10-2	1085_1	TTccctttatttCC
1086	tttccctttatttc	55498	55511	2-10-2	1086_1	TTtccctttattTC
1087	atttccctttattt	55499	55512	2-10-2	1087_1	ATttccctttatTT
1088	tatttccctttatt	55500	55513	2-10-2	1088_1	TAtttccctttaTT
1089	gtatttccctttat	55501	55514	2-10-2	1089_1	GTatttccctttAT

Oligonucleotide compound represent specific designs of a motif sequence. Typically, capital letters represent beta-D-oxy LNA nucleosides, lowercase letters represent DNA nucleosides, all LNA C are 5-methyl cytosine, and 5-methyl DNA cytosines are presented by “e” or ^{m}c, all internucleoside linkages are phosphorothioate internucleoside linkages.
Design refers to the gapmer design, F-G-F’, where each number represents the number of consecutive modified nucleosides, e.g. 2’ modified nucleosides (first number=5’ flank), followed by the number of DNA nucleosides (second number= gap region), followed by the number of modified nucleosides, e.g. 2’ modified nucleosides (third number=3’ flank), optionally preceded by or followed by further repeated regions of DNA and LNA, which are not necessarily part of the contiguous nucleotide sequence that is complementary to the target nucleic acid.
Motif sequences represent the contiguous sequence of nucleobases present in the oligonucleotide, also referred to as the Oligonucleotide Base Sequence.

The invention provides antisense oligonucleotides according to the invention, such as antisense oligonucleotides 12 – 24, such as 12 – 18 in length, nucleosides in length wherein the antisense oligonucleotide comprises a contiguous nucleotide sequence comprising at least 12, such as at least 14, such as at least 15 contiguous nucleotides present in a of sequence selected from SEQ ID NO 4 to SEQ ID NO: 1089; SEQ ID Nos 1099 to 1127; and SEQ ID NO 1137 – 1988.

DCM: Dichloromethane

DMF: Dimethylformamide

DMT: 4,4'-Dimethoxytrityl

THF: Tetrahydrofurane

Bz: Benzoyl

Ibu: Isobutyryl

RP-HPLC: Reverse phase high performance liquid chromatography

T_m Assay:

Oligonucleotide and RNA target (phosphate linked, PO) duplexes are diluted to 3 mM in 500 ml RNase-free water and mixed with 500 ml 2x T_m-buffer (200mM NaCl, 0.2mM EDTA, 20mM Naphosphate, pH 7.0). The solution is heated to 95ºC for 3 min and then allowed to anneal in room temperature for 30 min. The duplex melting temperatures (T_m) is measured on a Lambda 40 UV/VIS Spectrophotometer equipped with a Peltier temperature programmer PTP6 using PE Templab software (Perkin Elmer). The temperature is ramped up from 20ºC to 95ºC and then down to 25ºC, recording absorption at 260 nm. First derivative and the local maximums of both the melting and annealing are used to assess the duplex T_m.

Cell lines

Table 3: Details in relation to the cell lines for assaying the compounds:

Cell lines				Cells/well (96 well plate)	Hours of cell incubation prior to treatment	Days of treatment
Name	**Vendor**	**Cat.no.**	**Cell medium**			
A431	ECACC	85090402	EMEM (Cat.no. M2279), 10% FBS (Cat.no. F7524), 2mM Glutamine (Cat.no. G8541), 0.1 mM NEAA (Cat.no. M7145), 25µg/ml Gentamicin (Cat.no. G1397)	8000	24	3
NCI-H23	ATCC	CRL-5800	RPMI 1640 (Cat.no. R2405), 10% FBS (Cat.no. F7524),	10000	24	3

Cell lines				Cells/well (96 well plate)	Hours of cell incubation prior to treatment	Days of treatment
Name	Vendor	Cat.no.	Cell medium			
			10mM Hepes (Cat.no. H0887), 1mM Sodium Pyruvate (Cat.no. S8636), 25µg/ml Gentamicin (Cat.no. G1397)			
ARPE19	ATCC	CRL-2302	DMEM/F-12 HAM (Cat.no. D8437), 10% FBS (Cat.no. F7524), 25µg/ml Gentamicin (Cat.no. G1397)	2000	0	4
U251	ECACC	9063001	EMEM (Cat.no. M2279), 10% FBS (Cat.no. F7524), 2mM Glutamine (Cat.no. G8541), 0.1 mM NEAA (Cat.no. M7145), 1mM Sodium Pyruvate (Cat.no. S8636), 25µg/ml Gentamicin (Cat.no. G1397)	2000	0	4
U2-OS	ATCC	HTB-96	MCCoy 5A medium (Cat.no. M8403), 10% FBS (Cat.no. F7524), 1.5mM Glutamine (Cat.no. G8541), 25µg/ml Gentamicin	7000	24	3

Cell lines				Cells/well (96 well plate)	Hours of cell incubation prior to treatment	Days of treatment
Name	Vendor	Cat.no.	Cell medium			
			(Cat.no. G1397)			
SK-N-AS	ATCC	CRL-2137	Dulbecco's Modified Eagle's Medium, supplemented with 0.1 mM Non-Essential Amino Acids (NEAA) and fetal bovine serum to a final concentration of 10%	9300	24	4
iCell GlutaNeurons	Stemcell Technologies	R1034	BrainPhys Neuronal Medium (Cat.no. 5790) supplemented with iCell GlutaNeurons Kit (Stemcell Technoligies. no. R1034) according to vendor), N-2 (Thermo Fisher), 1µg/ml Laminin 512 (BioLamina, no. LN521)	50.000	168	4

* All medium and additives are purchased from Sigma Aldrich unless otherweise stated.

Example 1 Testing in vitro efficacy of LNA oligonucleotides in SK-N-AS, A431, NCI-H23 and ARPE19 cell lines at 25 and 5µM

An oligonucleotide screen is performed in human cell lines using the LNA oligonucleotides in table 2 (CMP ID NO: 4_1 - 1089_1, see column "oligonucleotide compounds") targeting SEQ ID NO 1. The human cell lines SK-N-AS, A341, NCI-H23 and ARPE19 are purchased from the vendors listed in table 3, and are maintained as recommended by the supplier in a humidified incubator at 37°C with 5% CO_2. For the screening assays, cells are seeded in 96

Oligonucleotide compound represent specific designs of a motif sequence. Capital letters represent beta-D-oxy LNA nucleosides, lowercase letters represent DNA nucleosides, all LNA C are 5-methyl cytosine, all internucleoside linkages are phosphorothioate internucleoside linkages.

Table 4: Compound and Data Table

SEQID	CMPID	Oligonucleotide Base Sequence	Oligonucleotide compound	% of ATXN3 mRNA remaining
1099	1099_1	CCAAAAGAAACCAAACCC	CCAAaagaaaccaaacCC	90,6
1100	1100_1	CCCCATTCAAATATTTATT	CCccattcaaatatttATT	90,5
1101	1101_1	AATCATTTACCCCCAAC	AAtcatttaccccCAAC	92
1102	1102_1	TATCTCAAACTATCCCCA	TAtctcaaactatcccCA	93
1103	1103_1	TCTATTCCTTAACCCAAC	TCTattccttaacccAAC	76,6
1104	1104_1	TCCCCTAAATAATTTAATCA	TCccctaaataatttaATCA	79,3
1105	1105_1	AAACCACTCCATTCCAA	AaaccactccattCCAA	57,7
1106	1106_1	TCTAAACCCCAAACTTTCA	TCtaaaccccaaactttCA	74,3
1107	1107_1	TTCTAAACCCCAAACTTTC	TTCtaaaccccaaacttTC	61,8
1108	1108_1	AGTTCTAAACCCCAAACT	AGttctaaaccccaaACT	73,7
1109	1109_1	TGAAACCATTACTACAACC	TGaaaccattactacAACC	24,9
1110	1110_1	ACATCATTTATCACTACCAC	ACAtcatttatcactaccAC	71,9
1111	1111_1	AACATTAAACCCTCCCA	AacattaaaccctcCCA	80,2
1112	1112_1	TCAGATCCTAAAATCACT	TCAgatcctaaaatcACT	79,5
1113	1113_1	CTATACCTAAAACAATCTA	CTAtacctaaaacaatCTA	99,1
1114	1114_1	TGATTCTTATACTTACTA	TGAttcttatacttaCTA	72,1
1115	1115_1	TAAAAATATAACTACTCCTA	TAaaaatataactactCCTA	93,7
1116	1116_1	TCTTCATTATACCATCAAAT	TCTtcattataccatcaAAT	51,5
1117	1117_1	GTTTCATATTTTTAATCC	GTTtcatatttttaaTCC	37,7
1118	1118_1	TAATATCCTCATTACCCATT	TAatatcctcattacccaTT	84
1119	1119_1	CAAATATTCACAAATCCTA	CAaatattcacaaatCCTA	73,3
1120	1120_1	CATCACAAAATAACCTATCA	CATcacaaaataacctaTCA	79,9
1121	1121_1	CTCTCAACTTCTACTACTAA	CTCtcaacttctactactAA	59,6
1122	1122_1	AATCTTATTTACATCTTCC	AATcttatttacatctTCC	20,7
1123	1123_1	CCAAAATTACTTCTTTTATC	CCAaaattacttcttttATC	56,5
1124	1124_1	AACCCAACTTTCTATTTT	AACCcaactttctattTT	52,7
1125	1125_1	ACAATATATTCCTCAATCA	ACAatatattcctcaaTCA	86,8
1126	1126_1	CCTGTAACAATTATACA	CCTgtaacaattatACA	92,3
1127	1127_1	CATCCCTTTACCACTTT	CAtccctttaccactTT	94,5

In the oligonucleotide compound column, capital letters represent beta-D-oxy LNA nucleosides, LNA cytosines are 5-methyl cytosine, lower case letters are DNA nucleosides, and all internucleoside linkages are phosphorothioate.

As can be seen, most of the above compounds targeting the listed target sequence regions are capable of inhibiting the expression of the human ataxin 3 transcript and that compounds targeting the target sequence region complementary to SEQ ID Nos 1122 and 1109 are particularly effective in inhibiting the human ataxin 3 transcript. Other effective target sites for ATXN3 can be determined from the above table.

Example 3

The screening assay described in Example 2 was performed using a series of further oligonucleotide targeting human ATXN3 pre-mRNA using the qpCR: (ATXN3_exon_8-9(1) PrimeTime® XL qPCR Assay (IDT).

qPCR probe and primers set 2:
Probe: 5'-/56-FAM/CTCCGCAGG/ZEN/GCT ATTCAGCT AAGT /31ABkFQ/-3' (SEQ ID NO 1134)
Primer 1: 5'-AGT AAGATTTGT ACCTGATGTCTGT-3' (SEQ ID NO 1135)
Primer 2: 5'-CATGGAAGATGAGGAAGCAGAT-3' (SEQ ID NO 1136)
The results are shown in the following table

Table 5

SEQID	CMPID	Oligonucleotide Base Sequence	Oligonucleotide compound	% of ATXN3 mRNA remaining
1137	1137_1	CCTACTTCACTTCCTAA	CctacttcacttcCTAA	68,9
1138	1138_1	TTTCCTACTTCACTTCCTA	TttcctacttcacttccTA	95,1
1139	1139_1	TTCCTACTTCACTTCCTA	TtcctacttcacttcCTA	85
1140	1140_1	TTTCCTACTTCACTTCCT	TTtcctacttcacttcCT	88,1
1141	1141_1	TTTCCTACTTCACTTCC	TttcctacttcactTCC	83,1
1142	1142_1	GTTTCCTACTTCACTTC	GTTtcctacttcactTC	60,2
1143	1143_1	ACCAAACCCAAACATCCC	AccaaacccaaacatcCC	88
1144	1144_1	AGAAACCAAACCCAAACATC	AgaaaccaaacccaaaCATC	91,3
1145	1145_1	AGAAACCAAACCCAAACAT	AGaaaccaaacccaaACAT	93,5
1146	1146_1	CTCCTAATACCTAAAAACAAA	CTCCtaatacctaaaaacaAA	100
1147	1147_1	CTCCTAATACCTAAAAACA	CTCCtaatacctaaaaaCA	94,2
1148	1148_1	ACTCCTAATACCTAAAAACA	ACTCctaatacctaaaaaCA	81
1149	1149_1	CACTCCTAATACCTAAAAACA	CACtcctaatacctaaaaACA	90,4
1150	1150_1	CCACTCCTAATACCTAAAAA	CCACtcctaatacctaaaAA	63
1151	1151_1	TCCACTCCTAATACCTAAAAA	TCCactcctaatacctaaaAA	54
1152	1152_1	CCACTCCTAATACCTAAAA	CCACtcctaatacctaaAA	73,7
1153	1153_1	TCCACTCCTAATACCTAAAA	TCCactcctaatacctaAAA	59
1154	1154_1	CCACTCCTAATACCTAAA	CCACtcctaatacctaAA	65,2
1155	1155_1	GTCCACTCCTAATACCTAAA	GtccactcctaataccTAAA	86,8

1156	1156_1	CCACTCCTAATACCTAA	CCactcctaatacCTAA	52,3
1157	1157_1	TCCACTCCTAATACCTAA	TCcactcctaatacCTAA	64,3
1157	1157_2	TCCACTCCTAATACCTAA	TCCActcctaatacctAA	66
1158	1158_1	GTCCACTCCTAATACCTAA	GtccactcctaataccTAA	85,5
1159	1159_1	AGTCCACTCCTAATACCTA	AgtccactcctaataccTA	87,4
1160	1160_1	TCCACTCCTAATACCTA	TCcactcctaatacCTA	70,1
1161	1161_1	AGTCCACTCCTAATACCT	AgtccactcctaatacCT	84,2
1162	1162_1	GTCCACTCCTAATACC	GTCcactcctaataCC	57,8
1163	1163_1	AGTCCACTCCTAATACC	AGtccactcctaataCC	77,1
1164	1164_1	CAGTCCACTCCTAATACC	CagtccactcctaatACC	86,7
1162	1162_2	GTCCACTCCTAATACC	GTCcactcctaatACC	67,8
1165	1165_1	CCAGTCCACTCCTAATAC	CcagtccactcctaaTAC	85,4
1166	1166_1	CAGTCCACTCCTAATAC	CAgtccactcctaaTAC	60,7
1167	1167_1	AGTCCACTCCTAATAC	AGTCcactcctaatAC	78,9
1168	1168_1	CAGTCCACTCCTAATA	CAGtccactcctaaTA	44,5
1169	1169_1	CCAGTCCACTCCTAATA	CCagtccactcctaaTA	33,8
1170	1170_1	GCAACTCTTTCCAAACA	GCAActctttccaaaCA	36
1171	1171_1	AGCAACTCTTTCCAAACA	AGCaactctttccaaaCA	35,3
1172	1172_1	CAGCAACTCTTTCCAAACA	CAgcaactctttccaaACA	58,3
1173	1173_1	CCAGCAACTCTTTCCAAA	CcagcaactctttcCAAA	69,7
1174	1174_1	CCAGCAACTCTTTCCAA	CCagcaactctttcCAA	42,1
1175	1175_1	ACCAGCAACTCTTTCCAA	ACcagcaactctttcCAA	65
1176	1176_1	TTACCAGCAACTCTTTC	TTACcagcaactcttTC	53
1177	1177_1	TGCTCCTCCTATTAAATAA	TGCtcctcctattaaatAA	76,3
1178	1178_1	GCTCCTCCTATTAAATAA	GCtcctcctattaaATAA	61,8
1179	1179_1	GCTCCTCCTATTAAATA	GCtcctcctattaaATA	60,2
1180	1180_1	TGCTCCTCCTATTAAATA	TGctcctcctattaaATA	70,2
1181	1181_1	TGCTCCTCCTATTAAAT	TGCtcctcctattaaAT	80,2
1182	1182_1	TTGCTCCTCCTATTAAAT	TTGCtcctcctattaaAT	79
1183	1183_1	ATTTAATAAAACAAAAACCCT	ATttaataaaacaaaaaCCCT	97,2
1184	1184_1	GCCCAAAAAACTAAATT	GCCCaaaaaactaaaTT	95,5
1185	1185_1	GTTTTTACATTCTAACTT	GTTtttacattctaaCTT	54,1
1186	1186_1	TGTTTTTACATTCTAACT	TGTTtttacattctaaCT	63,8
1187	1187_1	CTGTTTTTACATTCTAAC	CTGTttttacattctaAC	62,5
1188	1188_1	CCCCATTCAAATATTTAT	CCCcattcaaatattTAT	64,9
1189	1189_1	GCCCCATTCAAATATTTAT	GCcccattcaaatattTAT	86,2
1188	1188_2	CCCCATTCAAATATTTAT	CCCCattcaaatatttAT	96,2
1190	1190_1	GCCCCATTCAAATATTTA	GCcccattcaaatatTTA	82,2
1191	1191_1	CCATTCAAATATATACATTTT	CCATtcaaatatatacattTT	72
1191	1191_2	CCATTCAAATATATACATTTT	CCATtcaaataTatacattTT	37,7
1192	1192_1	TCCATTCAAATATATACATTT	TCCAttcaaatatatacatTT	56,8
1193	1193_1	ATCCATTCAAATATATACATT	ATCCattcaaaTatatacaTT	48
1194	1194_1	TCCATTCAAATATATACATT	TCCAttcaaatatatacaTT	53,7
1193	1193_2	ATCCATTCAAATATATACATT	ATCCattcaaatatatacaTT	54,7

1195	1195_1	TATCCATTCAAATATATACAT	TATccattcaaatatataCAT	80,1
1196	1196_1	TCCATTCAAATATATACAT	TCCattcaaatatataCAT	43,1
1197	1197_1	ATCCATTCAAATATATACA	ATCCattcaaatatataCA	53,9
1198	1198_1	TTATCCATTCAAATATATACA	TTatccattcaaatataTACA	69,4
1199	1199_1	TCCATTCAAATATATACA	TCCAttcaaatatataCA	54,7
1200	1200_1	TATCCATTCAAATATATACA	TATCcattcaaatatataCA	53,3
1201	1201_1	CTTTATCCATTCAAATATATA	CTttatccattcaaataTATA	85,5
1202	1202_1	TCTTTATCCATTCAAATATAT	TCTttatccattcaaataTAT	62,6
1203	1203_1	CTCTTTATCCATTCAAATATA	CTCtttatccattcaaatATA	38,4
1204	1204_1	TCTTTATCCATTCAAATATA	TCtttatccattcaaaTATA	70,9
1203	1203_2	CTCTTTATCCATTCAAATATA	CTCTttatccattcaaataTA	33,6
1205	1205_1	CTTTATCCATTCAAATATA	CTttatccattcaaaTATA	78,4
1206	1206_1	TCTCTTTATCCATTCAAATAT	TCtctttatccattcaaaTAT	82
1207	1207_1	CTCTTTATCCATTCAAATAT	CTCtttatccattcaaaTAT	39,8
1208	1208_1	TCTTTATCCATTCAAATAT	TCTttatccattcaaaTAT	63,1
1209	1209_1	TCTCTTTATCCATTCAAATA	TCtctttatccattcaAATA	65,2
1210	1210_1	CTCTTTATCCATTCAAATA	CTCTttatccattcaaaTA	32,2
1211	1211_1	TCTCTTTATCCATTCAAAT	TCTCtttatccattcaaAT	42
1212	1212_1	TCTCTTTATCCATTCAAA	TCTCtttatccattcaAA	42,5
1213	1213_1	AGCACCATATATATCTCA	AgcaccatatatatCTCA	16
1214	1214_1	GCACCATATATATCTCA	GCaccatatatatcTCA	16
1215	1215_1	CAGCACCATATATATCTCA	CagcaccatatatatCTCA	19,2
1215	1215_2	CAGCACCATATATATCTCA	CAgcaccatatatatcTCA	24,1
1216	1216_1	AGCACCATATATATCTC	AGCaccatatatatcTC	19,9
1217	1217_1	GCACCATATATATCTC	GCACcatatatatcTC	82,7
1218	1218_1	CAGCACCATATATATCTC	CAgcaccatatatatCTC	21,1
1219	1219_1	CAGCACCATATATATCT	CAGcaccatatataTCT	28,9
1220	1220_1	ACAGCACCATATATATCT	ACAGcaccatatatatCT	21,9
1221	1221_1	ACAGCACCATATATATC	ACAGcaccatatataTC	25,4
1222	1222_1	CTATGTTATTATCCCCA	CTAtgttattatcccCA	56,1
1223	1223_1	TCTATGTTATTATCCCC	TctatgttattatcCCC	47,7
1224	1224_1	CTCTACACTCTAACTCT	CtctacactctaaCTCT	79,3
1225	1225_1	TCTCTACACTCTAACTCT	TctctacactctaaCTCT	79,6
1226	1226_1	TTCTCTACACTCTAACTCT	TTCtctacactctaactCT	86,9
1227	1227_1	CTTCTCTACACTCTAACTCT	CTtctctacactctaactCT	97
1227	1227_2	CTTCTCTACACTCTAACTCT	CttctctacactctaacTCT	84,5
1228	1228_1	TTCTCTACACTCTAACTC	TTCtctacactctaacTC	81,4
1229	1229_1	CTTCTCTACACTCTAACTC	CttctctacactctaaCTC	89,1
1230	1230_1	TCTCTACACTCTAACTC	TCtctacactctaaCTC	87,3
1231	1231_1	CCTTCTCTACACTCTAACTC	CcttctctacactctaacTC	98,3
1232	1232_1	TTCTCTACACTCTAACT	TTCTctacactctaaCT	80,1
1233	1233_1	CTTCTCTACACTCTAACT	CTTctctacactctaaCT	73,6
1234	1234_1	CCTTCTCTACACTCTAACT	CcttctctacactctAACT	77,7
1235	1235_1	CCTTCTCTACACTCTAAC	CCTtctctacactctaAC	82,4

1236	1236_1	CTTCTCTACACTCTAAC	CTtctctacactcTAAC	90,6
1237	1237_1	AGCCTTCTCTACACTCTAA	AgccttctctacactcTAA	80
1238	1238_1	CCTTCTCTACACTCTAA	CCttctctacactcTAA	72,2
1239	1239_1	GCCTTCTCTACACTCTAA	GCcttctctacactctAA	62,9
1240	1240_1	AGCCTTCTCTACACTCTA	AgccttctctacactcTA	85,2
1241	1241_1	TACTAACTACAACACAAATCA	TACtaactacaacacaaaTCA	91,3
1241	1241_2	TACTAACTACAACACAAATCA	TACtaactacAacacaaaTCA	81,1
1242	1242_1	CTACTAACTACAACACAAATC	CTACtaactacaacacaaaTC	108
1243	1243_1	CACTACTAACTACAACACAAA	CACTactaactacaacacaAA	74
1244	1244_1	CACTACTAACTACAACACAA	CACtactaactacaacaCAA	87,4
1245	1245_1	ACACTACTAACTACAACACAA	ACActactaactacaacaCAA	84,1
1246	1246_1	CACTACTAACTACAACACA	CACTactaactacaacaCA	83,5
1247	1247_1	ACACTACTAACTACAACACA	ACACtactaactacaacaCA	81,3
1248	1248_1	GACACTACTAACTACAACAC	GACactactaactacaaCAC	51,6
1249	1249_1	GACACTACTAACTACAACA	GACActactaactacaaCA	51
1250	1250_1	AGACACTACTAACTACAA	AGAcactactaactaCAA	57,2
1251	1251_1	AGACACTACTAACTACA	AGACactactaactaCA	34,7
1252	1252_1	ATCATTTACCCCCAACCT	AtcatttacccccaacCT	96
1253	1253_1	ATCATTTACCCCCAACC	AtcatttacccccAACC	89,1
1254	1254_1	CAAATCATTTACCCCCAA	CaaatcatttacccCCAA	92
1255	1255_1	CCAAATCATTTACCCCCAA	CcaaatcatttaccccCAA	91
1256	1256_1	ACCAAATCATTTACCCCCA	AccaaatcatttaccccCA	89,9
1257	1257_1	TACCAAATCATTTACCCCC	TaccaaatcatttacccCC	84
1258	1258_1	ACCAAATCATTTACCCC	ACcaaatcatttacCCC	69,9
1259	1259_1	TACCAAATCATTTACCCC	TACcaaatcatttaccCC	56,3
1260	1260_1	CTACCAAATCATTTACCCC	CtaccaaatcatttaccCC	94
1261	1261_1	TACCAAATCATTTACCC	TAccaaatcatttACCC	68,9
1262	1262_1	CTACCAAATCATTTACC	CTACcaaatcatttaCC	70,3
1263	1263_1	TGCTACCAAATCATTTACC	TgctaccaaatcattTACC	79
1264	1264_1	GCTACCAAATCATTTACC	GCtaccaaatcatttACC	83,6
1265	1265_1	TGCTACCAAATCATTTAC	TGCTaccaaatcatttAC	88,3
1266	1266_1	GCTACCAAATCATTTAC	GCTaccaaatcattTAC	71,4
1267	1267_1	TGCTACCAAATCATTTA	TGCtaccaaatcatTTA	79,8
1268	1268_1	CTGCTACCAAATCATTTA	CTGctaccaaatcatTTA	75,3
1269	1269_1	ACTGCTACCAAATCATTT	ACTGctaccaaatcatTT	83,4
1270	1270_1	CTGCTACCAAATCATTT	CTGCtaccaaatcatTT	83
1271	1271_1	ACTGCTACCAAATCATT	ACTGctaccaaatcaTT	71,1
1272	1272_1	CACTTTGCCATAATCAA	CActttgccataaTCAA	26
1273	1273_1	TTATCTCAAACTATCCCCA	TTAtctcaaactatcccCA	92,9
1274	1274_1	ATCTCAAACTATCCCCA	ATctcaaactatccCCA	72,3
1275	1275_1	CTTATCTCAAACTATCCCCA	CttatctcaaactatcccCA	85,5
1276	1276_1	TATCTCAAACTATCCCC	TatctcaaactatcCCC	79,8
1277	1277_1	CTTATCTCAAACTATCCCC	CTtatctcaaactatccCC	84
1278	1278_1	TTATCTCAAACTATCCCC	TTAtctcaaactatccCC	89,7

1279	1279_1	CTTATCTCAAACTATCCC	CttatctcaaactaTCCC	83,5
1280	1280_1	CCTTATCTCAAACTATCCC	CcttatctcaaactatcCC	87,6
1279	1279_2	CTTATCTCAAACTATCCC	CTtatctcaaactatCCC	76,9
1281	1281_1	TTATCTCAAACTATCCC	TtatctcaaactaTCCC	84,7
1282	1282_1	CTTATCTCAAACTATCC	CTTatctcaaactaTCC	78,3
1283	1283_1	CCCTTATCTCAAACTATCC	CccttatctcaaactaTCC	76,4
1284	1284_1	CCTTATCTCAAACTATCC	CCTtatctcaaactatCC	69,3
1285	1285_1	CCTTATCTCAAACTATC	CCttatctcaaacTATC	75,9
1286	1286_1	GCCCTTATCTCAAACTATC	GCccttatctcaaactaTC	76,6
1287	1287_1	CCCTTATCTCAAACTATC	CCcttatctcaaacTATC	67,2
1288	1288_1	TGCCCTTATCTCAAACTAT	TgcccttatctcaaacTAT	90,5
1289	1289_1	GCCCTTATCTCAAACTAT	GCccttatctcaaacTAT	71,9
1290	1290_1	CCCTTATCTCAAACTAT	CCCTtatctcaaactAT	77,7
1291	1291_1	GCCCTTATCTCAAACTA	GCccttatctcaaacTA	68,4
1292	1292_1	TGCCCTTATCTCAAACTA	TgcccttatctcaaaCTA	81,5
1293	1293_1	TGCCCTTATCTCAAACT	TGcccttatctcaaaCT	75,7
1294	1294_1	TTGCCCTTATCTCAAAC	TTGCccttatctcaaAC	89
1295	1295_1	CTTGCCCTTATCTCAA	CTtgcccttatcTCAA	48,2
1296	1296_1	TGAAATCAAACTTCATCA	TGAaatcaaacttcaTCA	66,5
1297	1297_1	GGTCACCATACTTAAT	GGTCaccatacttaAT	89,7
1298	1298_1	TGCTAACACAAATTTCCT	TGctaacacaaattTCCT	47,3
1299	1299_1	GCTAACACAAATTTCCT	GCTaacacaaatttCCT	48,9
1299	1299_2	GCTAACACAAATTTCCT	GCtaacacaaattTCCT	45,7
1300	1300_1	TTGCTAACACAAATTTCC	TTGCtaacacaaatttCC	60,7
1301	1301_1	TGCTAACACAAATTTCC	TGCTaacacaaatttCC	62,6
1302	1302_1	TTGCTAACACAAATTTC	TTGCtaacacaaattTC	72,4
1303	1303_1	CCTTTGCTAACACAAAT	CCTTtgctaacacaaAT	48,1
1304	1304_1	GTATAACCAATAATAACTA	GTAtaaccaataataaCTA	86,1
1305	1305_1	TCTGACATCACACAATTT	TCTGacatcacacaatTT	67,8
1306	1306_1	TCTGACATCACACAATT	TCTGacatcacacaaTT	70,2
1307	1307_1	TATCTGACATCACACAA	TATctgacatcacaCAA	69,8
1308	1308_1	CTATTCCTTAACCCAAC	CTattccttaaccCAAC	77,7
1309	1309_1	GTCTATTCCTTAACCCAAC	GtctattccttaaccCAAC	86,2
1310	1310_1	GTCTATTCCTTAACCCAA	GtctattccttaacCCAA	60,4
1311	1311_1	TCTATTCCTTAACCCAA	TCTattccttaaccCAA	51
1312	1312_1	GTCTATTCCTTAACCCA	GtctattccttaacCCA	67,3
1313	1313_1	GTCTATTCCTTAACCC	GtctattccttaaCCC	77,4
1314	1314_1	GGTCTATTCCTTAACC	GGtctattccttaaCC	83,2
1315	1315_1	AGAACATTTCCTTCTCCT	AgaacatttccttctcCT	84,2
1316	1316_1	AACTGTCCCAAACAACC	AACtgtcccaaacaaCC	75
1317	1317_1	TTAGTCTCCCTCATTTTC	TtagtctccctcattTTC	72,4
1318	1318_1	ATTTAGTCTCCCTCATT	ATttagtctccctCATT	48
1319	1319_1	ATGCATCAAATCTCATA	ATGCatcaaatctcaTA	83,7
1320	1320_1	CCTAAATAATTTAATCATTAA	CCTaaataatttaatcatTAA	94

1321	1321_1	CCCTAAATAATTTAATCATTA	CCCTaaataatttaatcatTA	88,8
1322	1322_1	CCCCTAAATAATTTAATCATT	CCCctaaataatttaatcATT	80,7
1323	1323_1	CCCTAAATAATTTAATCATT	CCCTaaataatttaatcaTT	82,2
1324	1324_1	CCCTAAATAATTTAATCAT	CCCtaaataatttaatCAT	87,1
1325	1325_1	CCCCTAAATAATTTAATCA	CCCctaaataatttaaTCA	79,9
1326	1326_1	CCCTAAATAATTTAATCA	CCCtaaataatttaaTCA	82,5
1327	1327_1	CCCCTAAATAATTTAATC	CCCCtaaataatttaaTC	116
1328	1328_1	TCCCCTAAATAATTTAATC	TCCCctaaataatttaaTC	109
1329	1329_1	TTGCTAATATTTCCAAAA	TTGCtaatatttccaaAA	84,2
1330	1330_1	CTTGCTAATATTTCCAA	CTtgctaatatttCCAA	66,1
1331	1331_1	ACTGTCATCCATATTTCC	ActgtcatccatattTCC	66,2
1332	1332_1	ACTGTCATCCATATTTC	ACTgtcatccatatTTC	48,1
1333	1333_1	AATGCCCCACTCTAATAT	AATGccccactctaatAT	36,9
1334	1334_1	TGCCCCACTCTAATAT	TGCcccactctaatAT	52,8
1335	1335_1	ATGCCCCACTCTAATAT	ATgccccactctaaTAT	43,7
1336	1336_1	AAATGCCCCACTCTAATA	AAATgccccactctaaTA	25,7
1337	1337_1	ATGCCCCACTCTAATA	ATGccccactctaaTA	28,6
1338	1338_1	AATGCCCCACTCTAATA	AATGccccactctaaTA	29,9
1339	1339_1	TTAAATGCCCCACTCTA	TtaaatgccccactCTA	51,3
1340	1340_1	TCTGAAAATTCACTATCT	TCTGaaaattcactatCT	35,7
1341	1341_1	GTCTACTATATACATCT	GTCtactatatacaTCT	30,6
1342	1342_1	AGTCTACTATATACATCT	AGTCtactatatacatCT	45,3
1343	1343_1	AGTCTACTATATACATC	AGTCtactatatacaTC	57
1344	1344_1	GTCTACTATATACATC	GTCTactatatacaTC	46,5
1345	1345_1	TAGTCTACTATATACATC	TAgtctactatataCATC	68,3
1346	1346_1	TAGTCTACTATATACAT	TAGtctactatataCAT	89
1347	1347_1	CTAGTCTACTATATACAT	CTAgtctactatataCAT	86,6
1348	1348_1	CTAGTCTACTATATACA	CTAGtctactatataCA	88,5
1349	1349_1	ACTAGTCTACTATATAC	ACTagtctactataTAC	85,1
1350	1350_1	CTAGTCTACTATATAC	CTAgtctactataTAC	85,3
1351	1351_1	GTATATTCTACCCATAA	GTAtattctacccaTAA	51,3
1352	1352_1	TGTATATTCTACCCATAA	TGTatattctacccaTAA	48,4
1353	1353_1	TGTATATTCTACCCATA	TGtatattctaccCATA	45,6
1354	1354_1	ATGTATATTCTACCCATA	ATgtatattctaccCATA	90,2
1355	1355_1	ATGTATATTCTACCCAT	ATgtatattctacCCAT	51,1
1356	1356_1	GAAAACCACACAATTCCTA	GaaaaccacacaattCCTA	58,9
1357	1357_1	GAAAACCACACAATTCCT	GAaaaccacacaatTCCT	56,4
1358	1358_1	AGAAAACCACACAATTCCT	AGaaaaccacacaattCCT	58,4
1359	1359_1	CAGAAAACCACACAATTCC	CAGaaaaccacacaatTCC	43,3
1360	1360_1	AGAAAACCACACAATTCC	AGAaaaccacacaatTCC	47,6
1361	1361_1	CCAGAAAACCACACAATTC	CCAGaaaaccacacaatTC	26,3
1362	1362_1	CCAGAAAACCACACAATT	CCAGaaaaccacacaaTT	21
1363	1363_1	TCCAGAAAACCACACAAT	TCCAgaaaaccacacaAT	47,1
1364	1364_1	TTCCAGAAAACCACACAA	TTCCagaaaaccacacAA	49,8

1364	1364_2	TTCCAGAAAACCACACAA	TTCcagaaaaccacaCAA	45,8
1365	1365_1	GATATATCACTAAATCCAT	GAtatatcactaaatCCAT	27,4
1366	1366_1	GATATATCACTAAATCCA	GAtatatcactaaaTCCA	43,7
1367	1367_1	AGATATATCACTAAATCCA	AGatatatcactaaaTCCA	37,4
1368	1368_1	AGATATATCACTAAATCC	AGAtatatcactaaaTCC	33,6
1369	1369_1	TCATATATAAATTTCTCTA	TCAtatataaatttctCTA	78
1369	1369_2	TCATATATAAATTTCTCTA	TCATatataaatttctcTA	73,1
1370	1370_1	TCATATATAAATTTCTCT	TCatatataaatttCTCT	27,9
1370	1370_2	TCATATATAAATTTCTCT	TCATatataaatttctCT	60,2
1371	1371_1	AAGATCACACAACCATA	AAGAtcacacaaccaTA	19,8
1372	1372_1	TAAAAGATCACACAACCA	TAaaagatcacacaACCA	47,5
1373	1373_1	CATCACATAAAAACCCACTT	CATcacataaaaacccaCTT	45,4
1374	1374_1	CATCACATAAAAACCCACT	CAtcacataaaaaccCACT	57,9
1375	1375_1	TCATCACATAAAAACCCACT	TCatcacataaaaaccCACT	30,1
1376	1376_1	CATCACATAAAAACCCAC	CAtcacataaaaacCCAC	61,6
1377	1377_1	TCATCACATAAAAACCCAC	TCatcacataaaaacCCAC	30,6
1378	1378_1	GTCATCACATAAAAACCCAC	GTCatcacataaaaaccCAC	24,9
1379	1379_1	GTCATCACATAAAAACCCA	GTCatcacataaaaacCCA	28,7
1380	1380_1	TCATCACATAAAAACCCA	TCatcacataaaaaCCCA	43,9
1381	1381_1	CATCACATAAAAACCCA	CAtcacataaaaaCCCA	71,5
1382	1382_1	TCATCACATAAAAACCC	TCAtcacataaaaaCCC	42,9
1383	1383_1	GTCATCACATAAAAACCC	GTCatcacataaaaaCCC	24,9
1384	1384_1	AGTCATCACATAAAAACCC	AGtcatcacataaaaACCC	35,8
1384	1384_2	AGTCATCACATAAAAACCC	AGTCatcacataaaaacCC	23
1385	1385_1	TAGTCATCACATAAAAACC	TAGTcatcacataaaaaCC	36,3
1386	1386_1	AGTCATCACATAAAAACC	AGTCatcacataaaaaCC	34,9
1387	1387_1	ATGCTAAATACAAATCT	ATGCtaaatacaaatCT	81
1388	1388_1	GAAACCATTACTACAACCAA	GAaaccattactacaaCCAA	20,1
1389	1389_1	GAAACCATTACTACAACCA	GAaaccattactacaACCA	15,9
1390	1390_1	ATGAAACCATTACTACAAC	ATGAaaccattactacaAC	45,6
1391	1391_1	CATGAAACCATTACTACA	CATGaaaccattactaCA	55,9
1392	1392_1	CCATGAAACCATTACTAC	CCatgaaaccattaCTAC	29,5
1393	1393_1	CTCCCATGAAACCATTA	CTCCcatgaaaccatTA	73,7
1394	1394_1	TGCTTACTTTATACAAAA	TGCTtactttatacaaAA	55,9
1395	1395_1	ATGTTAATACTTTTTCCA	ATGttaatactttttCCA	92,9
1396	1396_1	CCTAATTTAACCCACAA	CCTaatttaacccaCAA	32,2
1397	1397_1	ATCCTAATTTAACCCACAA	ATCctaatttaacccaCAA	38,1
1398	1398_1	TCCTAATTTAACCCACAA	TCCtaatttaacccaCAA	39,9
1399	1399_1	TAATCCTAATTTAACCCACAA	TAAtcctaatttaacccaCAA	72,8
1400	1400_1	TAATCCTAATTTAACCCACA	TAatcctaatttaaccCACA	45
1401	1401_1	AATCCTAATTTAACCCACA	AATCctaatttaacccaCA	41,2
1402	1402_1	TCCTAATTTAACCCACA	TCCtaatttaacccACA	38,3
1403	1403_1	TAATCCTAATTTAACCCAC	TAatcctaatttaacCCAC	37,5
1404	1404_1	ATCCTAATTTAACCCAC	ATcctaatttaacCCAC	34,4

1405	1405_1	AATCCTAATTTAACCCAC	AAtcctaatttaacCCAC	48,2
1406	1406_1	TAATCCTAATTTAACCCA	TAatcctaatttaaCCCA	56,5
1407	1407_1	AATCCTAATTTAACCCA	AAtcctaatttaaCCCA	71,7
1408	1408_1	GTAATCCTAATTTAACCCA	GtaatcctaatttaaCCCA	63,6
1409	1409_1	TAATCCTAATTTAACCC	TAatcctaatttaACCC	56,5
1410	1410_1	GTAATCCTAATTTAACCC	GTaatcctaatttaACCC	44
1411	1411_1	AGTAATCCTAATTTAACCC	AGtaatcctaatttaaCCC	66,2
1410	1410_2	GTAATCCTAATTTAACCC	GTAatcctaatttaaCCC	34,2
1412	1412_1	AGTAATCCTAATTTAACC	AGTAatcctaatttaaCC	42,7
1413	1413_1	TCATTTATCACTACCACA	TCAtttatcactaccACA	26,5
1414	1414_1	CATCATTTATCACTACCACA	CatcatttatcactacCACA	46
1415	1415_1	CATTTATCACTACCACA	CAtttatcactacCACA	19,4
1416	1416_1	ATCATTTATCACTACCACA	ATcatttatcactacCACA	16,8
1416	1416_2	ATCATTTATCACTACCACA	ATCAtttatcactaccaCA	14,1
1417	1417_1	ACATCATTTATCACTACCACA	ACatcatttatcactaccACA	53,4
1418	1418_1	TCATTTATCACTACCAC	TCAtttatcactacCAC	18,9
1419	1419_1	ATCATTTATCACTACCAC	ATcatttatcactaCCAC	21,8
1420	1420_1	CATCATTTATCACTACCAC	CATcatttatcactacCAC	25,1
1421	1421_1	AACATCATTTATCACTACCAC	AACatcatttatcactacCAC	30,5
1421	1421_2	AACATCATTTATCACTACCAC	AacatcatttatcactaCCAC	40,4
1420	1420_2	CATCATTTATCACTACCAC	CatcatttatcactaCCAC	34,3
1422	1422_1	AACATCATTTATCACTACCA	AAcatcatttatcactACCA	34
1423	1423_1	CATCATTTATCACTACCA	CATCatttatcactacCA	11,3
1424	1424_1	TAACATCATTTATCACTACCA	TAacatcatttatcactaCCA	63,1
1425	1425_1	ACATCATTTATCACTACCA	ACATcatttatcactacCA	19
1422	1422_2	AACATCATTTATCACTACCA	AACatcatttatcactaCCA	25
1424	1424_2	TAACATCATTTATCACTACCA	TaacatcatttatcactACCA	61,3
1425	1425_2	ACATCATTTATCACTACCA	ACatcatttatcactACCA	23,5
1426	1426_1	TAACATCATTTATCACTACC	TAACatcatttatcactaCC	33,6
1427	1427_1	ACATCATTTATCACTACC	ACatcatttatcacTACC	32,3
1428	1428_1	TTAACATCATTTATCACTACC	TTAacatcatttatcactaCC	75,5
1429	1429_1	AACATCATTTATCACTACC	AACAtcatttatcactaCC	37,3
1430	1430_1	TTAACATCATTTATCACTAC	TTaacatcatttatcaCTAC	69,1
1431	1431_1	TAACATCATTTATCACTAC	TAacatcatttatcaCTAC	66,6
1432	1432_1	ATTAACATCATTTATCACTAC	ATtaacatcatttatcaCTAC	84,2
1432	1432_2	ATTAACATCATTTATCACTAC	ATtaacatcAtttatcaCTAC	62,8
1433	1433_1	ATTAACATCATTTATCACTA	ATTaacatcatttatcaCTA	81,3
1434	1434_1	TTAACATCATTTATCACTA	TTAacatcatttatcaCTA	74,5
1435	1435_1	TAATTAACATCATTTATCACT	TAattaacatcatttatCACT	84,3
1435	1435_2	TAATTAACATCATTTATCACT	TAattaacaTcatttatCACT	43,3
1436	1436_1	CTAATTAACATCATTTATCAC	CTaattaacatcatttaTCAC	81,4
1436	1436_2	CTAATTAACATCATTTATCAC	CTaattaacAtcatttaTCAC	46,7
1437	1437_1	CCTAATTAACATCATTTATCA	CCtaattaacatcatttaTCA	93,8
1438	1438_1	CTAATTAACATCATTTATCA	CTAattaacatcatttaTCA	89,6

1439	1439_1	CCTAATTAACATCATTTATC	CCTAattaacatcatttaTC	69,4
1440	1440_1	CCCTAATTAACATCATTTATC	CCctaattaacatcatttATC	86,3
1441	1441_1	CCTAATTAACATCATTTAT	CCTaattaacatcattTAT	87,4
1442	1442_1	CCTAATTAACATCATTTA	CCTAattaacatcattTA	66
1443	1443_1	CCCTAATTAACATCATTTA	CCCtaattaacatcattTA	88,7
1444	1444_1	GCCCTAATTAACATCATTT	GCCctaattaacatcatTT	87,9
1445	1445_1	CCCTAATTAACATCATTT	CCCTaattaacatcatTT	75,6
1446	1446_1	CGGCCCTAATTAACAT	CGGCcctaattaacAT	103
1447	1447_1	CTCGGCCCTAATTAA	CTCggccctaatTAA	57,4
1448	1448_1	CACATATAACATATAAACACA	CACAtataacatataaacaCA	61,7
1449	1449_1	TCACATATAACATATAAACAC	TCAcatataacatataaaCAC	43,6
1450	1450_1	ACTATCACATATAACATATA	ACTAtcacatataacataTA	58,5
1451	1451_1	CACTATCACATATAACATATA	CACTatcacatataacataTA	28,1
1452	1452_1	CACTATCACATATAACATAT	CACtatcacatataacaTAT	52
1453	1453_1	CACTATCACATATAACATA	CACTatcacatataacaTA	24,3
1454	1454_1	CACTATCACATATAACAT	CACtatcacatataaCAT	40,1
1455	1455_1	CACTATCACATATAACA	CACTatcacatataaCA	27
1456	1456_1	CAAAGTTTTCCCATTAC	CAaagttttcccaTTAC	21
1457	1457_1	ACAAAGTTTTCCCATTA	ACAaagttttcccaTTA	20,5
1458	1458_1	TCAGTCCAACATAACTC	TCAGtccaacataacTC	15,2
1459	1459_1	CAGTCCAACATAACTC	CAGtccaacataaCTC	23,5
1460	1460_1	ATCAGTCCAACATAACTC	ATCAgtccaacataacTC	13,7
1461	1461_1	ATCAGTCCAACATAACT	ATCAgtccaacataaCT	15,9
1462	1462_1	TAAACATTAAACCCTCCCAAA	TAaacattaaaccctccCAAA	87
1463	1463_1	AACATTAAACCCTCCCAA	AAcattaaaccctcCCAA	68,4
1464	1464_1	TAAACATTAAACCCTCCCAA	TaaacattaaaccctcCCAA	79,2
1465	1465_1	AAACATTAAACCCTCCCAA	AAacattaaaccctcCCAA	70,8
1466	1466_1	TATAAACATTAAACCCTCCCA	TAtaaacattaaaccctccCA	94
1467	1467_1	AACATTAAACCCTCCC	AAcattaaacccTCCC	78,3
1468	1468_1	AAACATTAAACCCTCCC	AAacattaaacccTCCC	89,4
1469	1469_1	TAAACATTAAACCCTCCC	TAAacattaaaccctCCC	72,9
1470	1470_1	ATAAACATTAAACCCTCCC	AtaaacattaaacccTCCC	86
1471	1471_1	TATAAACATTAAACCCTCC	TAtaaacattaaaccCTCC	91,1
1472	1472_1	TAAACATTAAACCCTCC	TAaacattaaaccCTCC	82,2
1473	1473_1	ACTATAAACATTAAACCCTCC	ActataaacattaaaccCTCC	86,5
1474	1474_1	ATAAACATTAAACCCTCC	ATaaacattaaaccCTCC	88,4
1475	1475_1	AACTATAAACATTAAACCCTC	AActataaacattaaacCCTC	92,6
1476	1476_1	CTATAAACATTAAACCCTC	CTataaacattaaacCCTC	82,9
1477	1477_1	ACTATAAACATTAAACCCTC	ACtataaacattaaacCCTC	89,8
1478	1478_1	AAACTATAAACATTAAACCCT	AAactataaacattaaaCCCT	98,9
1479	1479_1	CTATAAACATTAAACCCT	CTataaacattaaaCCCT	82,2
1480	1480_1	ACTATAAACATTAAACCCT	ACtataaacattaaaCCCT	86,6
1481	1481_1	AACTATAAACATTAAACCCT	AActataaacattaaaCCCT	89,5
1482	1482_1	GCTTTAAACTATAAACATT	GCtttaaactataaaCATT	58,2

1483	1483_1	TGCTTTAAACTATAAACA	TGCTttaaactataaaCA	57,2
1484	1484_1	CAGATTTATCACTATTA	CAGAtttatcactatTA	15,4
1485	1485_1	TCACAGCCTATCACCAC	TCacagcctatcacCAC	47,3
1485	1485_2	TCACAGCCTATCACCAC	TCAcagcctatcaccAC	46,3
1486	1486_1	ATCACAGCCTATCACCA	AtcacagcctatcACCA	56,9
1486	1486_2	ATCACAGCCTATCACCA	ATCacagcctatcacCA	23,7
1487	1487_1	AATCACAGCCTATCACC	AATCacagcctatcaCC	32,9
1487	1487_2	AATCACAGCCTATCACC	AatcacagcctatCACC	52,2
1488	1488_1	ATCACAGCCTATCACC	AtcacagcctatCACC	60,1
1489	1489_1	GCGTCACCCAAATCAC	GCgtcacccaaatCAC	11
1490	1490_1	AGCGTCACCCAAATCA	AG^{m}cgtcacccaaaTCA	17,4
1491	1491_1	AGCGTCACCCAAATC	AG^{m}cgtcacccaAATC	18,8
1492	1492_1	CAGATCCTAAAATCACT	CAGAtcctaaaatcaCT	71,8
1493	1493_1	TCAGATCCTAAAATCAC	TCAgatcctaaaatCAC	66,2
1494	1494_1	AGTAAAACCAATCATCAT	AGTaaaaccaatcatCAT	30,8
1495	1495_1	AGTAAAACCAATCATCA	AGTaaaaccaatcaTCA	24,2
1496	1496_1	CCCTTCCATCTCTACTAAAA	CccttccatctctactaaAA	89,7
1497	1497_1	ATAACTACATAACAAACCCA	ATaactacataacaaaCCCA	69,1
1498	1498_1	AATAACTACATAACAAACCCA	AAtaactacataacaaaCCCA	77,8
1499	1499_1	AACTACATAACAAACCCA	AActacataacaaaCCCA	62,9
1500	1500_1	TAACTACATAACAAACCCA	TAactacataacaaaCCCA	65
1501	1501_1	ACTACATAACAAACCCA	ACtacataacaaaCCCA	60,4
1502	1502_1	CAATAACTACATAACAAACCC	CAAtaactacataacaaaCCC	72,6
1503	1503_1	ATAACTACATAACAAACCC	ATAactacataacaaaCCC	60,2
1504	1504_1	ACAATAACTACATAACAAACC	ACAataactacataacaaACC	78,5
1504	1504_2	ACAATAACTACATAACAAACC	ACAAtaactacataacaaaCC	80,9
1505	1505_1	TGAATTCACAATAACTACA	TGaattcacaataacTACA	38,1
1506	1506_1	GCACATTTTTCTTAAACT	GCACatttttcttaaaCT	62,2
1507	1507_1	GCTATACCTAAAACAATCT	GCTatacctaaaacaaTCT	62,2
1508	1508_1	GCTATACCTAAAACAATC	GCTAtacctaaaacaaTC	68,9
1509	1509_1	CCCTTGTAACTAAAAAT	CCCTtgtaactaaaaAT	100
1510	1510_1	CCCCTTGTAACTAAAAA	CCCCttgtaactaaaAA	86,1
1511	1511_1	CCCCTTGTAACTAAAA	CCCCttgtaactaaAA	101
1512	1512_1	ACCCCTTGTAACTAAA	ACCCcttgtaactaAA	88,8
1513	1513_1	CACCCCTTGTAACTAA	CAccccttgtaaCTAA	80,4
1514	1514_1	ACACCCCTTGTAACTA	ACACcccttgtaacTA	72,4
1515	1515_1	GCTAAAACTAATCATCT	GCTaaaactaatcaTCT	72,2
1516	1516_1	GGCTAAAACTAATCAT	GGCtaaaactaatCAT	70,8
1517	1517_1	TTACCCTTCATATATACATCT	TtacccttcatatatacaTCT	89,4
1518	1518_1	ATTACCCTTCATATATACATC	AttacccttcatatataCATC	82,4
1519	1519_1	TTACCCTTCATATATACATC	TTAcccttcatatatacATC	56,3
1520	1520_1	CATTACCCTTCATATATACAT	CAttacccttcatatataCAT	84,2
1521	1521_1	TTACCCTTCATATATACAT	TTAcccttcatatataCAT	55,3
1522	1522_1	ATTACCCTTCATATATACAT	ATTacccttcatatataCAT	49,3

1523	1523_1	ACATTACCCTTCATATATACA	ACAttacccttcatatataCA	55,2
1523	1523_2	ACATTACCCTTCATATATACA	AcattacccttcatataTACA	63,4
1524	1524_1	TTACCCTTCATATATACA	TTACccttcatatataCA	46,9
1525	1525_1	CATTACCCTTCATATATACA	CattacccttcatataTACA	66
1526	1526_1	ATTACCCTTCATATATACA	ATTAcccttcatatataCA	36,7
1527	1527_1	ATTACCCTTCATATATAC	ATTacccttcatataTAC	46,6
1528	1528_1	TTACCCTTCATATATAC	TTAcccttcatataTAC	56,9
1529	1529_1	CATTACCCTTCATATATAC	CATtacccttcatataTAC	63,4
1530	1530_1	ACATTACCCTTCATATATAC	ACAttacccttcatataTAC	34,5
1531	1531_1	TACATTACCCTTCATATATAC	TAcattacccttcatataTAC	76,9
1532	1532_1	CATTACCCTTCATATATA	CAttacccttcataTATA	76,5
1533	1533_1	TACATTACCCTTCATATATA	TACattacccttcatatATA	36,5
1534	1534_1	ATTACCCTTCATATATA	ATtacccttcataTATA	78
1535	1535_1	ACATTACCCTTCATATATA	ACattacccttcataTATA	59,5
1536	1536_1	CATTACCCTTCATATAT	CATtacccttcataTAT	73,7
1537	1537_1	ACATTACCCTTCATATAT	ACAttacccttcataTAT	46,1
1538	1538_1	TACATTACCCTTCATATAT	TACattacccttcataTAT	36,9
1539	1539_1	ACATTACCCTTCATATA	ACattacccttcaTATA	54,2
1540	1540_1	TACATTACCCTTCATATA	TAcattacccttcaTATA	71,5
1541	1541_1	TACATTACCCTTCATAT	TACattacccttcaTAT	34,5
1542	1542_1	GATTCTTATACTTACTA	GATtcttatacttaCTA	46,2
1543	1543_1	TGATTCTTATACTTACT	TGattcttatactTACT	45,7
1544	1544_1	ATGATTCTTATACTTACT	ATGAttcttatacttaCT	54
1545	1545_1	GCCTCATTTTTACCTTT	GCctcatttttaccTTT	82,6
1546	1546_1	ACCAATCTTCTATTTTA	ACCAatcttctattTTA	94,8
1547	1547_1	CAACCAATCTTCTATTTTA	CAACcaatcttctatttTA	90,3
1548	1548_1	GCAACCAATCTTCTATTTT	GCAAccaatcttctattTT	88,3
1549	1549_1	GCAACCAATCTTCTATTT	GCAaccaatcttctaTTT	85
1550	1550_1	GCAACCAATCTTCTATT	GCaaccaatcttcTATT	87,3
1551	1551_1	TGCAACCAATCTTCTATT	TGCaaccaatcttctaTT	90,2
1552	1552_1	TAACTGCAACCAATCTT	TAactgcaaccaaTCTT	88,2
1553	1553_1	TGAATACAACACACATCA	TGAatacaacacacaTCA	97,4
1554	1554_1	ATGAATACAACACACATCA	ATGAatacaacacacatCA	84,4
1555	1555_1	TAAAAATATAACTACTCCT	TAaaaatataactacTCCT	99,8
1556	1556_1	GTAAAAATATAACTACTCC	GTaaaaatataactaCTCC	93,7
1557	1557_1	TCAACTGATACCCACAA	TCAactgatacccaCAA	57,7
1558	1558_1	TGTCTTAACATTTTTCTT	TGTCttaacatttttcTT	63,1
1559	1559_1	CCACTTCAAACTTTTAATTAA	CCActtcaaacttttaatTAA	85
1560	1560_1	CCACTTCAAACTTTTAATTA	CCACttcaaacttttaatTA	84,9
1561	1561_1	CCCACTTCAAACTTTTAATTA	CCcacttcaaacttttaaTTA	88,7
1562	1562_1	CCACTTCAAACTTTTAATT	CCACttcaaacttttaaTT	79,1
1563	1563_1	CCCACTTCAAACTTTTAATT	CCCacttcaaacttttaaTT	86,2
1564	1564_1	ACCCACTTCAAACTTTTAATT	ACCcacttcaaacttttaaTT	100
1565	1565_1	CCACTTCAAACTTTTAAT	CCACttcaaacttttaAT	85,3

1566	1566_1	ACCCACTTCAAACTTTTAAT	ACCcacttcaaacttttAAT	88,8
1567	1567_1	AACCCACTTCAAACTTTTAAT	AACCcacttcaaacttttaAT	92,3
1568	1568_1	CCCACTTCAAACTTTTAA	CCCacttcaaactttTAA	79,9
1569	1569_1	ACCCACTTCAAACTTTTAA	ACCcacttcaaactttTAA	82,5
1570	1570_1	CCCACTTCAAACTTTTA	CCCacttcaaactttTA	79,6
1571	1571_1	ACCCACTTCAAACTTTTA	ACCcacttcaaacttTTA	77,2
1572	1572_1	AACCCACTTCAAACTTTTA	AACCcacttcaaactttTA	86,2
1573	1573_1	ACCCACTTCAAACTTTT	ACCCacttcaaacttTT	93,3
1574	1574_1	AACCCACTTCAAACTTTT	AACCcacttcaaacttTT	82,7
1575	1575_1	AACCCACTTCAAACTTT	AACCcacttcaaactTT	85,8
1576	1576_1	GGACTCTATTAATCAA	GGActctattaatCAA	91,7
1577	1577_1	GAATATTCTACTCTTCT	GAatattctactcTTCT	95,3
1578	1578_1	CTGTATTTACCAATTCAA	CTGtatttaccaattCAA	90,8
1579	1579_1	CTGTATTTACCAATTCA	CTGTatttaccaattCA	88,7
1580	1580_1	ACTGTATTTACCAATTCA	ACTGtatttaccaattCA	97,3
1581	1581_1	ACTGTATTTACCAATTC	ACTGtatttaccaatTC	104
1582	1582_1	CACTGTATTTACCAATT	CACTgtatttaccaaTT	91,1
1583	1583_1	TCACTGTATTTACCAAT	TCACtgtatttaccaAT	98,6
1584	1584_1	CCAACTACTTTACTTTTCAAA	CCaactactttacttttCAAA	84,3
1585	1585_1	CCAACTACTTTACTTTTCAA	CCaactactttactttTCAA	80
1586	1586_1	ACCAACTACTTTACTTTTCAA	ACcaactactttactttTCAA	85,1
1585	1585_2	CCAACTACTTTACTTTTCAA	CCAactactttacttttCAA	75,2
1587	1587_1	CCAACTACTTTACTTTTCA	CCAactactttacttttCA	71,9
1588	1588_1	TACCAACTACTTTACTTTTCA	TaccaactactttacttTTCA	82,8
1587	1587_2	CCAACTACTTTACTTTTCA	CCAactactttactttTCA	67,7
1589	1589_1	ACCAACTACTTTACTTTTCA	ACcaactactttactttTCA	84
1590	1590_1	TACCAACTACTTTACTTTTC	TACcaactactttacttTTC	75,3
1591	1591_1	GTACCAACTACTTTACTTT	GTACcaactactttactTT	75,8
1592	1592_1	GTACCAACTACTTTACTT	GTAccaactactttaCTT	65,7
1593	1593_1	GTACCAACTACTTTACT	GTACcaactactttaCT	74,5
1594	1594_1	TGTACCAACTACTTTACT	TGtaccaactacttTACT	87,1
1595	1595_1	TTGTACCAACTACTTTAC	TTGtaccaactacttTAC	73,3
1596	1596_1	GTACCAACTACTTTAC	GTAccaactacttTAC	72,5
1597	1597_1	TGTACCAACTACTTTAC	TGTaccaactacttTAC	66
1598	1598_1	TTGTACCAACTACTTTA	TTGTaccaactacttTA	49,3
1599	1599_1	ATTTCATTTTTCTTTTAATA	ATTtcatttttcttttaATA	98,6
1599	1599_2	ATTTCATTTTTCTTTTAATA	ATTTcatttttcttttaaTA	90,7
1600	1600_1	CCTAATTTCATTTTTCTTTT	CCtaatttcatttttcTTTT	69,2
1601	1601_1	TCCTAATTTCATTTTTCTTT	TCctaatttcatttttCTTT	47
1602	1602_1	TTCTTCATTATACCATCAAAT	TTCTtcattataccatcaaAT	29,4
1603	1603_1	TTTCTTCATTATACCATCAAA	TTTCttcattataccatcaAA	24,1
1604	1604_1	TTTTCTTCATTATACCATCAA	TTttcttcattataccaTCAA	14,3
1605	1605_1	TCTTCATTATACCATCAA	TCttcattataccaTCAA	5,02
1606	1606_1	TTTCTTCATTATACCATCAA	TTtcttcattataccaTCAA	21,2

1607	1607_1	TTCTTCATTATACCATCAA	TTCttcattataccatCAA	5,83
1608	1608_1	ATATTTTCTTCATTATACCAT	AtattttcttcattataCCAT	76,1
1609	1609_1	ATATTTTCTTCATTATACCA	ATattttcttcattataCCA	40,2
1610	1610_1	AATATTTTCTTCATTATACCA	AATattttcttcattataCCA	37
1611	1611_1	AAATATTTTCTTCATTATACC	AAatattttcttcattaTACC	23,4
1612	1612_1	ATATTTTCTTCATTATACC	ATattttcttcattaTACC	14,2
1613	1613_1	AATATTTTCTTCATTATACC	AATAttttcttcattataCC	68
1614	1614_1	TAAATATTTTCTTCATTATA	TAaatattttcttcatTATA	96,8
1615	1615_1	TTTTCCTTCATCTACTTCT	TTTtccttcatctacttCT	42,8
1616	1616_1	ATTTTCCTTCATCTACTTCT	ATtttccttcatctacttCT	76
1617	1617_1	AATTTTCCTTCATCTACTTC	AATTttccttcatctactTC	54,9
1618	1618_1	AGAATTTTCCTTCATCTA	AgaattttccttcaTCTA	58
1619	1619_1	CAGAATTTTCCTTCATCT	CAgaattttccttcATCT	23,5
1620	1620_1	TCAGAATTTTCCTTCATC	TCAgaattttccttcaTC	29,7
1621	1621_1	CTAGAAATATCTCACATT	CTAGaaatatctcacaTT	64,6
1622	1622_1	CTAGAAATATCTCACAT	CTAgaaatatctcaCAT	75,5
1623	1623_1	ACTAGAAATATCTCACA	ACTAgaaatatctcaCA	53,2
1624	1624_1	ATTAGCCATTAATCTAT	ATtagccattaatCTAT	71,9
1625	1625_1	TTGTTACAAAATAATCCA	TTgttacaaaataaTCCA	12
1625	1625_2	TTGTTACAAAATAATCCA	TTGttacaaaataatCCA	23,8
1626	1626_1	TTATTTTTTACATTAACTA	TTAttttttacattaaCTA	92,1
1627	1627_1	TGCCAAAATACTAACATCA	TGCcaaaatactaacaTCA	32
1628	1628_1	GCCAAAATACTAACATCA	GCCaaaatactaacaTCA	27,8
1629	1629_1	TGCCAAAATACTAACATC	TGCCaaaatactaacaTC	61,5
1630	1630_1	GAGTACAACACTTACA	GAGTacaacacttaCA	31,8
1631	1631_1	CACATCCATTCATTTTAT	CACatccattcatttTAT	30,6
1632	1632_1	CCACATCCATTCATTTTAT	CCAcatccattcattttAT	21,7
1633	1633_1	CCACATCCATTCATTTTA	CCacatccattcattTTA	20
1634	1634_1	TATGCCACATCCATTCAT	TatgccacatccattCAT	47
1635	1635_1	TTATGCCACATCCATTCA	TtatgccacatccaTTCA	20,7
1636	1636_1	TATGCCACATCCATTCA	TAtgccacatccattCA	43,3
1637	1637_1	TTATGCCACATCCATTC	TtatgccacatccATTC	19,5
1638	1638_1	ATTATGCCACATCCATT	ATtatgccacatcCATT	25,1
1639	1639_1	AGTTTCATATTTTTAATC	AGTttcatatttttaATC	65,9
1640	1640_1	ATCACTGCACACTTTCC	ATCactgcacactttCC	12,9
1641	1641_1	AAGCTCTTTCCAAATTCT	AAGCtctttccaaattCT	34,6
1642	1642_1	TAGTTCTTAACTCTTCTC	TagttcttaactctTCTC	19,2
1643	1643_1	TTAGTTCTTAACTCTTC	TTAGttcttaactctTC	18
1644	1644_1	AGCTTCAAATACTCAAA	AGCTtcaaatactcaAA	74,5
1645	1645_1	TTTCAAAGCCACACCTA	TttcaaagccacaCCTA	66,9
1646	1646_1	AATATCCTCATTACCCATT	AATAtcctcattacccaTT	52,3
1647	1647_1	TATCCTCATTACCCATT	TAtcctcattaccCATT	53,4
1647	1647_2	TATCCTCATTACCCATT	TATCctcattacccaTT	22,3
1648	1648_1	ATATCCTCATTACCCATT	ATAtcctcattacccATT	55,8

1649	1649_1	AATATCCTCATTACCCAT	AAtatcctcattacCCAT	46,1
1650	1650_1	TAATATCCTCATTACCCAT	TAAtatcctcattaccCAT	58,3
1651	1651_1	TTAATATCCTCATTACCCAT	TTaatatcctcattaccCAT	61,8
1652	1652_1	ATATCCTCATTACCCAT	ATAtcctcattaccCAT	56,2
1653	1653_1	AATATCCTCATTACCCA	AAtatcctcattaCCCA	49,7
1654	1654_1	TAATATCCTCATTACCCA	TAATatcctcattaccCA	45,6
1655	1655_1	TTTAATATCCTCATTACCCA	TttaatatcctcattacCCA	67,5
1656	1656_1	TTAATATCCTCATTACCCA	TTaatatcctcattacCCA	36
1656	1656_2	TTAATATCCTCATTACCCA	TTAAtatcctcattaccCA	57,9
1654	1654_2	TAATATCCTCATTACCCA	TAAtatcctcattacCCA	40
1653	1653_2	AATATCCTCATTACCCA	AATatcctcattacCCA	44,8
1657	1657_1	ATTTAATATCCTCATTACCC	AtttaatatcctcattaCCC	59,9
1658	1658_1	TAATATCCTCATTACCC	TAATatcctcattacCC	32,9
1659	1659_1	TTAATATCCTCATTACCC	TTAAtatcctcattacCC	42
1660	1660_1	TTTAATATCCTCATTACCC	TttaatatcctcattACCC	41,1
1661	1661_1	AATTTAATATCCTCATTACCC	AatttaatatcctcattaCCC	61
1662	1662_1	TTTAATATCCTCATTACC	TTTAatatcctcattaCC	60,6
1663	1663_1	AATTTAATATCCTCATTACC	AAtttaatatcctcatTACC	58,8
1664	1664_1	TTAATATCCTCATTACC	TTaatatcctcatTACC	42,3
1665	1665_1	AAATTTAATATCCTCATTACC	AAatttaatatcctcatTACC	55,9
1666	1666_1	ATTTAATATCCTCATTACC	ATttaatatcctcatTACC	55,5
1667	1667_1	TAAATTTAATATCCTCATTAC	TAaatttaatatcctcaTTAC	78
1668	1668_1	TTAAATTTAATATCCTCATTA	TTAaatttaatatcctcaTTA	95,2
1669	1669_1	CTTAAATTTAATATCCTCATT	CTtaaatttaatatcctCATT	73,2
1670	1670_1	TCTTAAATTTAATATCCTCAT	TCttaaatttaatatccTCAT	46,8
1671	1671_1	TCTTAAATTTAATATCCTCA	TCttaaatttaatatcCTCA	29,8
1672	1672_1	TTCTTAAATTTAATATCCTCA	TTCttaaatttaatatccTCA	35
1673	1673_1	TTCTTAAATTTAATATCCTC	TTcttaaatttaatatCCTC	36,2
1674	1674_1	TCTTAAATTTAATATCCTC	TCttaaatttaatatCCTC	25,1
1675	1675_1	TTCTTAAATTTAATATCCT	TTCttaaatttaatatCCT	46,9
1676	1676_1	TCTTAAATTTAATATCCT	TCttaaatttaataTCCT	50,9
1677	1677_1	AATAGCCTTTATTCTAC	AAtagcctttattCTAC	33,6
1678	1678_1	CAGCAACAATTATTAATA	CAGCaacaattattaaTA	70,5
1679	1679_1	CCAGCAACAATTATTAAT	CCAGcaacaattattaAT	64,2
1680	1680_1	ACCAGCAACAATTATTAA	ACCagcaacaattatTAA	20,5
1680	1680_2	ACCAGCAACAATTATTAA	ACCAgcaacaattattAA	39,7
1681	1681_1	ACCAGCAACAATTATTA	ACCAgcaacaattatTA	39,4
1682	1682_1	TACCAGCAACAATTATT	TACCagcaacaattaTT	26,4
1683	1683_1	CCCCAAATCTAAAACACTTC	CCccaaatctaaaacacTTC	79,4
1684	1684_1	AACCCCAAATCTAAAACACTT	AACCccaaatctaaaacacTT	82
1685	1685_1	CCCCAAATCTAAAACACTT	CCCcaaatctaaaacacTT	86,4
1686	1686_1	AACCCCAAATCTAAAACACT	AACCccaaatctaaaacaCT	75,2
1687	1687_1	ACCCCAAATCTAAAACACT	ACcccaaatctaaaaCACT	72,5
1688	1688_1	ACCCCAAATCTAAAACAC	ACCccaaatctaaaaCAC	80,9

1689	1689_1	GCAAATATTCACAAATCCT	GCAaatattcacaaatCCT	20,7
1689	1689_2	GCAAATATTCACAAATCCT	GCaaatattcacaaaTCCT	29,3
1690	1690_1	ACTATTTAACACACATTATCA	ACTatttaacacacattaTCA	36,6
1691	1691_1	CTATTTAACACACATTATCA	CTAtttaacacacattaTCA	49,6
1692	1692_1	TACTATTTAACACACATTATC	TACTatttaacacacattaTC	52,4
1693	1693_1	ACTATTTAACACACATTATC	ACTAtttaacacacattaTC	51,8
1694	1694_1	TACTATTTAACACACATTAT	TACtatttaacacacatTAT	91,1
1695	1695_1	CTACTATTTAACACACATTAT	CTActatttaacacacatTAT	72,7
1696	1696_1	CTACTATTTAACACACATTA	CTACtatttaacacacatTA	47,4
1697	1697_1	ACTACTATTTAACACACATTA	ACTActatttaacacacatTA	38,3
1698	1698_1	CTACTATTTAACACACATT	CTACtatttaacacacaTT	41,6
1699	1699_1	ACTACTATTTAACACACATT	ACtactatttaacacaCATT	40,3
1700	1700_1	ACTACTATTTAACACACAT	ACTactatttaacacaCAT	36,8
1701	1701_1	CTACTATTTAACACACA	CTACtatttaacacaCA	45,9
1702	1702_1	ACTACTATTTAACACACA	ACTActatttaacacaCA	32,6
1703	1703_1	TATAGACCCTTAATATT	TATAgacccttaataTT	41,4
1704	1704_1	TTATAGACCCTTAATAT	TTAtagacccttaaTAT	68,5
1705	1705_1	CATCACAAAATAACCTATCAT	CAtcacaaaataacctaTCAT	86,8
1706	1706_1	TCATCACAAAATAACCTATCA	TCAtcacaaaataacctaTCA	67,4
1707	1707_1	TTCATCACAAAATAACCTATC	TTCAtcacaaaataacctaTC	49
1708	1708_1	TTCATCACAAAATAACCTA	TTcatcacaaaataaCCTA	76,4
1709	1709_1	TTTCATCACAAAATAACCTA	TTtcatcacaaaataaCCTA	88,6
1710	1710_1	TCATCACAAAATAACCTA	TCatcacaaaataaCCTA	59,2
1711	1711_1	TTTTCATCACAAAATAACCTA	TTttcatcacaaaataaCCTA	86,1
1712	1712_1	ATTTTCATCACAAAATAACCT	ATTttcatcacaaaataaCCT	64,8
1713	1713_1	TATTTTCATCACAAAATAACC	TATTttcatcacaaaataaCC	76,9
1713	1713_2	TATTTTCATCACAAAATAACC	TATTttcatcaCaaaataaCC	56
1714	1714_1	GTATTTTCATCACAAAATA	GTATtttcatcacaaaaTA	47
1715	1715_1	TTACCTAGATCACTCC	TtacctagatcaCTCC	73,1
1716	1716_1	CTTACCTAGATCACTC	CTTacctagatcaCTC	81,5
1717	1717_1	CCTTACCTAGATCACT	CCTtacctagatcaCT	95,9
1718	1718_1	TAACTGCTCCTTAATCC	TAActgctccttaatCC	34,8
1719	1719_1	TCTAGCAATCCTCTCCT	TCtagcaatcctctcCT	64,2
1720	1720_1	TTCTAGCAATCCTCTCC	TtctagcaatcctcTCC	70,4
1721	1721_1	TTTTCACCTACTAATATTCAT	TTttcacctactaatatTCAT	55,3
1722	1722_1	TTTCACCTACTAATATTCAT	TTtcacctactaatatTCAT	66,2
1723	1723_1	TTCACCTACTAATATTCAT	TTCacctactaatattCAT	17,2
1724	1724_1	TCACCTACTAATATTCAT	TCAcctactaatattCAT	23,5
1725	1725_1	TCACCTACTAATATTCA	TCAcctactaatatTCA	21,1
1726	1726_1	TTTCACCTACTAATATTCA	TTTCacctactaatattCA	16,7
1727	1727_1	TTTTCACCTACTAATATTCA	TTttcacctactaataTTCA	31,3
1728	1728_1	TTTTTCACCTACTAATATTCA	TTtttcacctactaataTTCA	45,3
1729	1729_1	TTCACCTACTAATATTCA	TTCAcctactaatattCA	24,7
1730	1730_1	ATTTTTCACCTACTAATATTC	ATTtttcacctactaataTTC	48,5

1731	1731_1	TTTTTCACCTACTAATATTC	TTTttcacctactaataTTC	31,5
1732	1732_1	TATTTTTCACCTACTAATATT	TAtttttcacctactaaTATT	90,2
1733	1733_1	TATTTTTCACCTACTAATAT	TATttttcacctactaaTAT	89,1
1734	1734_1	TTATTTTTCACCTACTAATAT	TTAtttttcacctactaaTAT	86,1
1735	1735_1	TTATTTTTCACCTACTAATA	TTATttttcacctactaaTA	52,9
1736	1736_1	TATTTTTCACCTACTAATA	TATTtttcacctactaaTA	54,9
1737	1737_1	TTTATTTTTCACCTACTAATA	TTTAtttttcacctactaaTA	52
1738	1738_1	TTTATTTTTCACCTACTAA	TTtatttttcacctaCTAA	51,2
1739	1739_1	TTTATTTTTCACCTACTA	TTTatttttcacctaCTA	19
1740	1740_1	CTCAACTTCTACTACTAATT	CTCAacttctactactaaTT	19,7
1741	1741_1	TCTCAACTTCTACTACTAATT	TCTCaacttctactactaaTT	25,8
1742	1742_1	CTCTCAACTTCTACTACTAAT	CTCtcaacttctactactAAT	43
1743	1743_1	CTCAACTTCTACTACTAAT	CTCAacttctactactaAT	20,1
1744	1744_1	TCTCAACTTCTACTACTAAT	TCTCaacttctactactaAT	22,8
1745	1745_1	TCTCTCAACTTCTACTACTAA	TCtctcaacttctactacTAA	58,4
1746	1746_1	CTCAACTTCTACTACTAA	CTcaacttctactaCTAA	47,3
1747	1747_1	TCTCAACTTCTACTACTAA	TCtcaacttctactaCTAA	56,3
1748	1748_1	CTCAACTTCTACTACTA	CTCaacttctactaCTA	10,7
1749	1749_1	TTCTCTCAACTTCTACTACTA	TtctctcaacttctactaCTA	79,1
1750	1750_1	TCTCTCAACTTCTACTACTA	TCtctcaacttctactacTA	61,2
1751	1751_1	TCTCAACTTCTACTACTA	TCtcaacttctactaCTA	66,8
1752	1752_1	CTCTCAACTTCTACTACTA	CtctcaacttctactACTA	61,7
1753	1753_1	CTCTCAACTTCTACTACT	CTCtcaacttctactaCT	37,9
1754	1754_1	TCTCAACTTCTACTACT	TCtcaacttctacTACT	51,1
1755	1755_1	TCTCTCAACTTCTACTACT	TCtctcaacttctactACT	44,2
1756	1756_1	TTTCTCTCAACTTCTACTACT	TTtctctcaacttctactACT	65,7
1757	1757_1	TTCTCTCAACTTCTACTACT	TTCtctcaacttctactaCT	33,5
1758	1758_1	TTTCTCTCAACTTCTACTAC	TTtctctcaacttctacTAC	67,9
1759	1759_1	CTCTCAACTTCTACTAC	CTCtcaacttctacTAC	34,1
1760	1760_1	TTCTCTCAACTTCTACTAC	TtctctcaacttctaCTAC	63,8
1761	1761_1	TTTTCTCTCAACTTCTACTAC	TTTTctctcaacttctactAC	20,6
1762	1762_1	TCTCTCAACTTCTACTAC	TCtctcaacttctacTAC	49,7
1763	1763_1	TTTCTCTCAACTTCTACTA	TTtctctcaacttctaCTA	60,2
1764	1764_1	TTTTCTCTCAACTTCTACTA	TtttctctcaacttctACTA	52,2
1765	1765_1	TTTTTCTCTCAACTTCTACTA	TTTttctctcaacttctacTA	40,2
1766	1766_1	TCTCTCAACTTCTACTA	TCtctcaacttctaCTA	47,5
1767	1767_1	TTCTCTCAACTTCTACTA	TTCtctcaacttctacTA	35,1
1768	1768_1	TTTCTCTCAACTTCTACT	TTTCtctcaacttctaCT	28,6
1769	1769_1	TTTTCTCTCAACTTCTACT	TTTtctctcaacttctaCT	44,1
1770	1770_1	CTTTTTCTCTCAACTTCTACT	CtttttctctcaacttctaCT	99,8
1771	1771_1	TTTTTCTCTCAACTTCTACT	TTTttctctcaacttctaCT	43,7
1772	1772_1	CTTTTTCTCTCAACTTCTAC	CTTtttctctcaacttctAC	36,2
1773	1773_1	ACTTTTTCTCTCAACTTCTAC	ACTttttctctcaacttctAC	35,6
1774	1774_1	TTTTCTCTCAACTTCTAC	TtttctctcaacttCTAC	38,6

1775	1775_1	TTTTTCTCTCAACTTCTAC	TttttctctcaacttCTAC	42,1
1776	1776_1	CTTTTTCTCTCAACTTCTA	CTttttctctcaacttCTA	41,2
1777	1777_1	TACTTTTTCTCTCAACTTCTA	TactttttctctcaacttCTA	69,4
1778	1778_1	ACTTTTTCTCTCAACTTCTA	ActttttctctcaacttCTA	66,2
1779	1779_1	TTTTTCTCTCAACTTCTA	TttttctctcaactTCTA	35,5
1780	1780_1	TACTTTTTCTCTCAACTTCT	TActttttctctcaacttCT	65
1781	1781_1	TTACTTTTTCTCTCAACTTCT	TtactttttctctcaactTCT	62,1
1782	1782_1	TTACTTTTTCTCTCAACTTC	TTActttttctctcaactTC	38,9
1783	1783_1	TACTTTTTCTCTCAACTTC	TACtttttctctcaactTC	34
1784	1784_1	ACTTTTTCTCTCAACTTC	ActttttctctcaaCTTC	19,7
1785	1785_1	TTACTTTTTCTCTCAACTT	TTActttttctctcaaCTT	22
1786	1786_1	TACTTTTTCTCTCAACTT	TACtttttctctcaaCTT	22,3
1787	1787_1	TTACTTTTTCTCTCAACT	TTACtttttctctcaaCT	11,6
1788	1788_1	GTTACTTTTTCTCTCAACT	GTtactttttctctcAACT	43,2
1789	1789_1	GTTACTTTTTCTCTCAAC	GTtactttttctctCAAC	29
1790	1790_1	GTTACTTTTTCTCTCAA	GTtactttttctcTCAA	5,53
1791	1791_1	AGTTACTTTTTCTCTCAA	AGTtactttttctctCAA	6,5
1792	1792_1	CTTTTACATTCCCATTAACA	CTTTtacattcccattaaCA	24,5
1793	1793_1	CACTTTTACATTCCCATTAAC	CACttttacattcccattaAC	25,3
1794	1794_1	CTTTTACATTCCCATTAAC	CTtttacattcccatTAAC	21,5
1795	1795_1	ACTTTTACATTCCCATTAAC	ACttttacattcccatTAAC	23
1796	1796_1	ACTTTTACATTCCCATTAA	ACttttacattcccaTTAA	30
1797	1797_1	CTTTTACATTCCCATTAA	CTtttacattcccaTTAA	27,4
1798	1798_1	CACTTTTACATTCCCATTAA	CActtttacattcccaTTAA	28
1798	1798_2	CACTTTTACATTCCCATTAA	CACttttacattcccatTAA	15,9
1799	1799_1	TACACTTTTACATTCCCATTA	TAcacttttacattcccatTA	52,2
1800	1800_1	ACTTTTACATTCCCATTA	ACTtttacattcccaTTA	13,1
1801	1801_1	CACTTTTACATTCCCATTA	CActtttacattcccATTA	15,7
1802	1802_1	ACACTTTTACATTCCCATTA	ACacttttacattcccaTTA	19,1
1802	1802_2	ACACTTTTACATTCCCATTA	ACActtttacattcccatTA	9,66
1803	1803_1	CACTTTTACATTCCCATT	CActtttacattccCATT	10,2
1804	1804_1	TACACTTTTACATTCCCATT	TACacttttacattcccaTT	10,3
1805	1805_1	ACACTTTTACATTCCCATT	ACACttttacattcccaTT	4,51
1805	1805_2	ACACTTTTACATTCCCATT	ACacttttacattccCATT	6,8
1806	1806_1	TACACTTTTACATTCCCAT	TACacttttacattccCAT	3,53
1806	1806_2	TACACTTTTACATTCCCAT	TACActtttacattcccAT	4,79
1807	1807_1	TACACTTTTACATTCCCA	TACacttttacattccCA	6,35
1808	1808_1	GTACACTTTTACATTCCCA	GtacacttttacattcCCA	3
1808	1808_2	GTACACTTTTACATTCCCA	GTacacttttacattccCA	16,3
1809	1809_1	GTACACTTTTACATTCCC	GTAcacttttacattcCC	4,33
1810	1810_1	TACACTTTTACATTCCC	TACacttttacattcCC	3,26
1811	1811_1	TGTACACTTTTACATTCCC	TGtacacttttacattcCC	12,3
1809	1809_2	GTACACTTTTACATTCCC	GtacacttttacattCCC	2,49
1812	1812_1	TGTACACTTTTACATTCC	TGtacacttttacatTCC	2,47

1813	1813_1	CTGTACACTTTTACATTC	CTGtacacttttacaTTC	1,89
1814	1814_1	ATCTTATTTACATCTTCC	ATcttatttacatcTTCC	5,41
1815	1815_1	GAATCTTATTTACATCTTC	GAatcttatttacatCTTC	25,8
1816	1816_1	GAATCTTATTTACATCTT	GAatcttatttacaTCTT	19,1
1817	1817_1	TGAATCTTATTTACATCT	TGAatcttatttacaTCT	41,3
1818	1818_1	ATTCAGCTTTTTCAATC	ATTCagctttttcaaTC	16,8
1819	1819_1	TTAATTTTCCCTTCACTCCT	TtaattttcccttcactcCT	85,8
1820	1820_1	TTAATTTTCCCTTCACTCC	TtaattttcccttcactCC	85,8
1821	1821_1	TTAATTTTCCCTTCACTC	TtaattttcccttcACTC	51
1822	1822_1	GTTAATTTTCCCTTCACTC	GttaattttcccttcACTC	27,2
1823	1823_1	CAAAATTACTTCTTTTATCAT	CAaaattacttcttttaTCAT	86,7
1823	1823_2	CAAAATTACTTCTTTTATCAT	CAaaattacTtcttttaTCAT	51,5
1824	1824_1	CCAAAATTACTTCTTTTATCA	CCAaaattacttcttttaTCA	31,3
1824	1824_2	CCAAAATTACTTCTTTTATCA	CCaaaattacttcttttATCA	36
1825	1825_1	TCCAAAATTACTTCTTTTATC	TCcaaaattacttcttttTATC	40,9
1826	1826_1	TCCAAAATTACTTCTTTTAT	TCCaaaattacttcttttTAT	50,2
1827	1827_1	CCAAAATTACTTCTTTTAT	CCAaaattacttcttttTAT	70
1828	1828_1	TTCCAAAATTACTTCTTTTAT	TTCcaaaattacttcttttTAT	64,9
1829	1829_1	TCCAAAATTACTTCTTTTA	TCCAaaattacttcttttTA	36,9
1830	1830_1	TTCCAAAATTACTTCTTTTA	TTCCaaaattacttcttttTA	52,2
1831	1831_1	GTTCCAAAATTACTTCTTT	GTTCcaaaattacttctTT	54,8
1832	1832_1	GTTCCAAAATTACTTCTT	GTtccaaaattactTCTT	12,5
1833	1833_1	TGTTCCAAAATTACTTCT	TGTtccaaaattactTCT	20,1
1834	1834_1	ATGTTCCAAAATTACTTC	ATGTtccaaaattactTC	23,8
1835	1835_1	CATATTTTACTCTTTTTATT	CATAttttactctttttaTT	90,6
1836	1836_1	CCATATTTTACTCTTTTTAT	CCATattttactctttttAT	35,4
1836	1836_2	CCATATTTTACTCTTTTTAT	CCAtattttactcttttTAT	60,8
1837	1837_1	CCCATATTTTACTCTTTTTAT	CccatattttactcttttTTAT	75,8
1838	1838_1	CATATTTTACTCTTTTTAT	CATattttactcttttTAT	83,2
1839	1839_1	CCCATATTTTACTCTTTTTA	CCcatattttactcttttTA	81,1
1840	1840_1	CCATATTTTACTCTTTTTA	CCatattttactcttTTTA	24,7
1841	1841_1	ACCCATATTTTACTCTTTTTA	AcccatattttactcttTTTA	59
1842	1842_1	CCATATTTTACTCTTTTT	CCATattttactcttttTT	21,6
1843	1843_1	CCCATATTTTACTCTTTTT	CCcatattttactcttTTT	77,2
1844	1844_1	ACCCATATTTTACTCTTTTT	ACccatattttactctTTTT	97,4
1845	1845_1	TACCCATATTTTACTCTTTTT	TAcccatattttactcttTTT	58,6
1846	1846_1	TACCCATATTTTACTCTTTT	TACccatattttactctTTT	20,4
1847	1847_1	CCCATATTTTACTCTTTT	CCCatattttactcttTT	93,2
1848	1848_1	ACCCATATTTTACTCTTTT	ACCcatattttactcttTT	21,8
1846	1846_2	TACCCATATTTTACTCTTTT	TAcccatattttactcTTTT	22,5
1849	1849_1	TTACCCATATTTTACTCTTTT	TTAcccatattttactcttTT	41,4
1850	1850_1	TACCCATATTTTACTCTTT	TAcccatattttactCTTT	18,9
1851	1851_1	ACCCATATTTTACTCTTT	ACCcatattttactcTTT	13,4
1852	1852_1	TTACCCATATTTTACTCTTT	TTacccatattttactCTTT	14,5

1853	1853_1	TTTACCCATATTTTACTCTTT	TTTacccatattttactcTTT	22,2
1852	1852_2	TTACCCATATTTTACTCTTT	TTACccatattttactctTT	16,7
1853	1853_2	TTTACCCATATTTTACTCTTT	TTTAcccatattttactctTT	16
1854	1854_1	TTACCCATATTTTACTCTT	TTAcccatattttactCTT	14
1855	1855_1	TTTACCCATATTTTACTCTT	TTtacccatattttacTCTT	14,9
1856	1856_1	ACCCATATTTTACTCTT	ACCcatattttactCTT	8,02
1857	1857_1	TACCCATATTTTACTCTT	TACccatattttactCTT	16,7
1858	1858_1	TACCCATATTTTACTCT	TACccatattttacTCT	22,3
1859	1859_1	TTACCCATATTTTACTCT	TTACccatattttactCT	15,2
1860	1860_1	TTTACCCATATTTTACTCT	TTTAcccatattttactCT	11,8
1861	1861_1	TTACCCATATTTTACTC	TTAcccatattttaCTC	24,4
1862	1862_1	TTTACCCATATTTTACTC	TTTacccatattttaCTC	14
1863	1863_1	GTTTACCCATATTTTACTC	GTttacccatattttaCTC	12,2
1864	1864_1	GTTTACCCATATTTTACT	GTttacccatatttTACT	24,9
1865	1865_1	TGTTTACCCATATTTTAC	TGTttacccatatttTAC	13,1
1866	1866_1	GTTTACCCATATTTTAC	GTttacccatattTTAC	13,2
1867	1867_1	TGTTTACCCATATTTTA	TGTttacccatattTTA	6,69
1868	1868_1	TTCTTGCTTCAACCATC	TtcttgcttcaacCATC	13,6
1869	1869_1	GTTACCTCCCTTTATAT	GTtacctccctttatAT	60,9
1870	1870_1	GGTTACCTCCCTTTAT	GgttacctccctTTAT	39
1871	1871_1	AGGTTACCTCCCTTTA	AggttacctcccTTTA	35,4
1872	1872_1	ATGTTCTCTATCTCTATA	ATGttctctatctctATA	53,3
1873	1873_1	TATGTTCTCTATCTCTA	TAtgttctctatctCTA	73,4
1874	1874_1	AGATCAAACTAAAACCT	AGAtcaaactaaaaCCT	88,7
1875	1875_1	TGCCCAATTTCACCCAA	TGcccaatttcacccAA	30,3
1876	1876_1	TTTGCCCAATTTCACCC	TttgcccaatttcacCC	53,3
1877	1877_1	TTTTGCCCAATTTCACC	TTttgcccaatttcaCC	57,8
1878	1878_1	TGTATATCAACAATTCAT	TGTatatcaacaattCAT	20,8
1879	1879_1	ACATTTCTTTAAAATTTCCA	ACatttctttaaaattTCCA	96,4
1879	1879_2	ACATTTCTTTAAAATTTCCA	ACAtttctttaaaatttCCA	96,6
1880	1880_1	CACATTTCTTTAAAATTTCCA	CACAtttctttaaaatttcCA	95,5
1879	1879_3	ACATTTCTTTAAAATTTCCA	AcatttctttaaaattTCCA	98,1
1879	1879_4	ACATTTCTTTAAAATTTCCA	ACATttctttaaaatttcCA	98
1881	1881_1	CCACATTTCTTTAAAATTTCC	CcacatttctttaaaatTTCC	90
1882	1882_1	CACATTTCTTTAAAATTTCC	CAcatttctttaaaattTCC	94,8
1882	1882_2	CACATTTCTTTAAAATTTCC	CAcatttctttaaaatTTCC	89,1
1882	1882_3	CACATTTCTTTAAAATTTCC	CACAtttctttaaaatttCC	94,4
1883	1883_1	ACATTTCTTTAAAATTTCC	ACAtttctttaaaattTCC	91,9
1882	1882_4	CACATTTCTTTAAAATTTCC	CACatttctttaaaattTCC	92,4
1884	1884_1	CCACATTTCTTTAAAATTTC	CCACatttctttaaaattTC	98,3
1885	1885_1	ACCACATTTCTTTAAAATTTC	ACCAcatttctttaaaattTC	97,5
1884	1884_2	CCACATTTCTTTAAAATTTC	CCAcatttctttaaaattTC	102
1884	1884_3	CCACATTTCTTTAAAATTTC	CCacatttctttaaaaTTTC	94,9
1884	1884_4	CCACATTTCTTTAAAATTTC	CCAcatttctttaaaatTTC	87,2

1886	1886_1	ACCACATTTCTTTAAAATTT	ACCAcatttctttaaaatTT	94,8
1887	1887_1	ACAAAACCACATTTCTTTAA	ACAaaaccacatttcttTAA	97,4
1888	1888_1	CTGTTTTCAAATCATTTC	CTGTtttcaaatcattTC	15,8
1889	1889_1	GAACCATTACTATTATCAA	GAaccattactattaTCAA	27,3
1890	1890_1	AGAACCATTACTATTATCA	AGAaccattactattaTCA	19,8
1891	1891_1	AGAACCATTACTATTATC	AGaaccattactatTATC	17,9
1892	1892_1	CTAGAACCATTACTATTA	CTAGaaccattactatTA	35,3
1893	1893_1	TAGAACCATTACTATTA	TAGAaccattactatTA	13,2
1894	1894_1	CTAGAACCATTACTATT	CTAGaaccattactaTT	32,1
1895	1895_1	AGATTACCATCTTTCAAAA	AGATtaccatctttcaaAA	59,5
1895	1895_2	AGATTACCATCTTTCAAAA	AGAttaccatctttcaAAA	54,1
1896	1896_1	AGATTACCATCTTTCAAA	AGATtaccatctttcaAA	50,6
1896	1896_2	AGATTACCATCTTTCAAA	AGattaccatctttCAAA	42,3
1897	1897_1	AGATTACCATCTTTCAA	AGAttaccatctttCAA	32,4
1898	1898_1	AAGATTACCATCTTTCA	AAGAttaccatctttCA	47,9
1899	1899_1	CATGCTCACACATTTTAA	CATgctcacacatttTAA	60,5
1899	1899_2	CATGCTCACACATTTTAA	CAtgctcacacattTTAA	70,3
1899	1899_3	CATGCTCACACATTTTAA	CAtgctcacacatttTAA	69,8
1899	1899_4	CATGCTCACACATTTTAA	CATGctcacacattttAA	55,9
1900	1900_1	CTTAAGCTATCTAAACA	CTTAagctatctaaaCA	82,6
1901	1901_1	TGAACAATTCAACATTCA	TGAacaattcaacatTCA	67,7
1902	1902_1	GATCAAAAAACTTTCCCT	GAtcaaaaaactttCCCT	76,1
1903	1903_1	AGATCAAAAAACTTTCCCT	AGatcaaaaaactttCCCT	70,4
1904	1904_1	AGATCAAAAAACTTTCCC	AGAtcaaaaaactttCCC	73,6
1905	1905_1	TCCTAGATCAAAAAACT	TCCTagatcaaaaaaCT	69,9
1906	1906_1	ATTTTTTCTTCTCTTTTCA	ATTTtttcttctcttttCA	8,98
1907	1907_1	TATTTTTTCTTCTCTTTTCA	TATtttttcttctcttttCA	63,8
1908	1908_1	ATATTTTTTCTTCTCTTTTC	ATatttttttcttctctTTTC	16,1
1909	1909_1	TCTGCTTTAAAAACTCTC	TCtgctttaaaaacTCTC	34,3
1910	1910_1	CTCTGCTTTAAAAACTC	CTCtgctttaaaaaCTC	51,6
1911	1911_1	ACTACACAAACACATTCAA	ACtacacaaacacatTCAA	37,6
1912	1912_1	CAAACTACACAAACACATTCA	CAaactacacaaacacaTTCA	41,2
1913	1913_1	ACAAACTACACAAACACATTC	ACAaactacacaaacacaTTC	63,1
1914	1914_1	CAACAAACTACACAAACACAT	CAAcaaactacacaaacaCAT	86,1
1915	1915_1	CACAACAAACTACACAAACAC	CACaacaaactacacaaaCAC	62,1
1916	1916_1	TCACAACAAACTACACAAACA	TCACaacaaactacacaaaCA	48,6
1917	1917_1	TTCACAACAAACTACACAAAC	TTCAcaacaaactacacaaAC	58,8
1918	1918_1	ATTTCACAACAAACTACACAA	ATTtcacaacaaactacaCAA	76,8
1919	1919_1	CAATTTCACAACAAACTACAC	CAAtttcacaacaaactaCAC	70,7
1920	1920_1	TGTAACAATTTCACAACAA	TGTaacaatttcacaaCAA	59,5
1921	1921_1	TGTAACAATTTCACAACA	TGTAacaatttcacaaCA	28,7
1922	1922_1	TTAAGCCAACCCCACCA	TtaagccaaccccacCA	83,1
1923	1923_1	TTTAAGCCAACCCCACC	TttaagccaaccccACC	69,2
1924	1924_1	ATTTAAGCCAACCCCAC	AtttaagccaaccCCAC	60,6

1925	1925_1	CCAGTAATACAAATTATA	CCAGtaatacaaattaTA	69,5
1926	1926_1	CCCAGTAATACAAATTA	CCCAgtaatacaaatTA	55,9
1927	1927_1	TCCCAGTAATACAAATT	TCCCagtaatacaaaTT	64,9
1928	1928_1	ATCCCAGTAATACAAAT	ATCCcagtaatacaaAT	65,9
1929	1929_1	CTACTAGCATCACCACT	CtactagcatcacCACT	19,8
1930	1930_1	TTCTACTAGCATCACC	TtctactagcatCACC	21,8
1931	1931_1	CTTCTACTAGCATCAC	CTtctactagcaTCAC	33,2
1932	1932_1	TAAATTACTCATTAAATCCAT	TAaattactcattaaatCCAT	77,8
1933	1933_1	ATAAATTACTCATTAAATCCA	ATaaattactcattaaaTCCA	52,4
1934	1934_1	TAAATTACTCATTAAATCCA	TAaattactcattaaaTCCA	51,6
1935	1935_1	CATAAATTACTCATTAAATCC	CATaaattactcattaaaTCC	58,5
1935	1935_2	CATAAATTACTCATTAAATCC	CATaaattacTcattaaaTCC	22,3
1936	1936_1	GATTTATTTTTCTACTTA	GAtttatttttctaCTTA	66
1937	1937_1	ATACAACAAACAATTCACTTT	ATacaacaaacaattcaCTTT	53,2
1937	1937_2	ATACAACAAACAATTCACTTT	ATACaacaaacaattcactTT	48,1
1938	1938_1	CGATACAACAAACAATTCA	CGATacaacaaacaattCA	23
1939	1939_1	GAACATCCACACTAACAACA	GAACatccacactaacaaCA	43,6
1940	1940_1	ACATCCACACTAACAACA	ACAtccacactaacaACA	65
1939	1939_2	GAACATCCACACTAACAACA	GAAcatccacactaacaACA	52
1939	1939_3	GAACATCCACACTAACAACA	GAacatccacactaacAACA	58,1
1941	1941_1	GAACATCCACACTAACAAC	GAACatccacactaacaAC	51,3
1941	1941_2	GAACATCCACACTAACAAC	GAacatccacactaaCAAC	63,3
1942	1942_1	TGAACATCCACACTAACAA	TGAacatccacactaaCAA	57,8
1943	1943_1	TTGAACATCCACACTAACA	TTGAacatccacactaaCA	60,3
1944	1944_1	TGAACATCCACACTAACA	TGAAcatccacactaaCA	42,6
1945	1945_1	CATTGAACATCCACACTA	CATtgaacatccacaCTA	59,4
1946	1946_1	ATTGAACATCCACACTA	ATTgaacatccacaCTA	50
1947	1947_1	CATTGAACATCCACACT	CAttgaacatccaCACT	43
1948	1948_1	ACTCATTGAACATCCAC	ACtcattgaacatCCAC	46,8
1949	1949_1	TATCTTTATTTAATAATCTT	TATCtttatttaataatcTT	93,4
1949	1949_2	TATCTTTATTTAATAATCTT	TAtctttatttaataaTCTT	96,9
1950	1950_1	TCTCAAGCTTCACTCTA	TCtcaagcttcactcTA	78,6
1951	1951_1	GACAATATATTCCTCAATC	GACAatatattcctcaaTC	73
1952	1952_1	GACAATATATTCCTCAAT	GACAatatattcctcaAT	82
1952	1952_2	GACAATATATTCCTCAAT	GAcaatatattcctCAAT	76,8
1953	1953_1	TCCTGTAACAATTATAC	TCCtgtaacaattaTAC	95,4
1954	1954_1	ACCCAGAATAAAAACCAC	ACccagaataaaaaCCAC	95,5
1955	1955_1	TTCCACTTTCTTACTCCC	TtccactttcttactcCC	96,6
1956	1956_1	TTCCACTTTCTTACTCC	TtccactttcttacTCC	86,3
1957	1957_1	TTTCCACTTTCTTACTCC	TttccactttcttacTCC	89,2
1958	1958_1	TTTCCACTTTCTTACTC	TTTCcactttcttacTC	89,2
1959	1959_1	ATCCCTTTACCACTTTT	ATCcctttaccactTTT	101
1960	1960_1	CATCCCTTTACCACTTTT	CAtccctttaccactTTT	98
1961	1961_1	TCATCCCTTTACCACTTT	TCatccctttaccactTT	101

1962	1962_1	TCATCCCTTTACCACTT	TCAtccctttaccacTT	96,9
1963	1963_1	CTCATCCCTTTACCACTT	CtcatccctttaccacTT	97,7
1964	1964_1	GTCTACATCTAACCCC	GtctacatctaacCCC	97
1965	1965_1	AGTCTACATCTAACCCC	AGtctacatctaaccCC	99,6
1966	1966_1	CAGTCTACATCTAACCCC	CagtctacatctaaccCC	97,4
1967	1967_1	CAGTCTACATCTAACCC	CagtctacatctaaCCC	99,5
1968	1968_1	TCAGTCTACATCTAACCC	TCagtctacatctaacCC	98,9
1969	1969_1	AGTCTACATCTAACCC	AGTctacatctaacCC	98,2
1970	1970_1	TCAGTCTACATCTAACC	TCagtctacatctAACC	98,3
1971	1971_1	TTCAGTCTACATCTAACC	TTCagtctacatctaaCC	98
1972	1972_1	TTCAGTCTACATCTAAC	TTCAgtctacatctaAC	98,7
1973	1973_1	TTTCAGTCTACATCTAA	TTtcagtctacatCTAA	90,1
1974	1974_1	AGTTTTAACCACACCTCCT	AgttttaaccacacctccCT	102
1975	1975_1	GTTTTAACCACACCTCC	GTTttaaccacacctCC	93,7
1976	1976_1	AGTTTTAACCACACCTCC	AgttttaaccacaccTCC	95
1977	1977_1	AGTTTTAACCACACCTC	AGttttaaccacacCTC	88,7
1978	1978_1	GAGTTTTAACCACACC	GAGttttaaccacACC	94,7
1979	1979_1	CAGATCTTCTCTTTATTT	CAGatcttctctttaTTT	96,3
1980	1980_1	TGTTTTCAACAAAACATCA	TGTtttcaacaaaacaTCA	89,9
1981	1981_1	TGTTTTCAACAAAACATC	TGttttcaacaaaaCATC	97,5
1982	1982_1	CTGTTTTCAACAAAACAT	CTGttttcaacaaaaCAT	102
1983	1983_1	TCTGTTTTCAACAAAACA	TCTGttttcaacaaaaCA	98
1984	1984_1	ATCTTTCTAAAACTTACC	ATCTttctaaaacttaCC	96,3
1985	1985_1	CAGAATCTTTCTAAAACT	CAGAatctttctaaaaCT	91,7
1986	1986_1	CTACAGAATCTTTCTAA	CTacagaatctttCTAA	97,6
1986	1986_2	CTACAGAATCTTTCTAA	CTAcagaatctttcTAA	95,6
1987	1987_1	ATTTCCCTTTATTTCCCTT	AtttccctttatttcccCTT	92
1988	1988_1	GTATTTCCCTTTATTTCC	GtatttccctttattTCC	99,5

In the oligonucleotide compound column, capital letters represent beta-D-oxy LNA nucleosides, LNA cytosines are 5-methyl cytosine, lower case letters are DNA nucleosides, and all internucleoside linkages are phosphorothioate. ^{m}c represent 5-methyl cytosine DNA nucleosides (used in compounds 1490_1 and 1491_1).

Example 4

The screening assay described in Example 2 was performed using a series of further oligonucleotide targeting human ATXN3 pre-mRNA using the qpCR: (ATXN3_exon_8-9(1) PrimeTime® XL qPCR Assay (IDT).

qPCR probe and primers set 2:

Probe: 5'-/56-FAM/CTCCGCAGG/ZEN/GCT ATTCAGCT AAGT /31ABkFQ/-3' (SEQ ID NO 1134)

Primer 1: 5'-AGT AAGATTTGT ACCTGATGTCTGT-3' (SEQ ID NO 1135)

Primer 2: 5'-CATGGAAGATGAGGAAGCAGAT-3' (SEQ ID NO 1136)

Table 6

SEQID	CMPID	Oligonucleotide Base Sequence	Oligonucleotide compound	% of ATXN3 mRNA remaining
1110	1110_2	ACATCATTTATCACTACCAC	ACatcatttatcactacCAC	44
1102	1102_2	TATCTCAAACTATCCCCA	TatctcaaactatccCCA	74
1104	1104_2	TCCCCTAAATAATTTAATCA	TCCcctaaataatttaaTCA	78
1116	1116_2	TCTTCATTATACCATCAAAT	TCTTcattataccatcaaAT	12
1121	1121_2	CTCTCAACTTCTACTACTAA	CtctcaacttctactaCTAA	68
1114	1114_2	TGATTCTTATACTTACTA	TGATtcttatacttacTA	64
1120	1120_2	CATCACAAAATAACCTATCA	CATCacaaaataacctatCA	38
1100	1100_2	CCCCATTCAAATATTTATT	CCCcattcaaatatttATT	79
1112	1112_2	TCAGATCCTAAAATCACT	TCAGatcctaaaatcaCT	65
1123	1123_2	CCAAAATTACTTCTTTTATC	CCaaaattacttctttTATC	37
1117	1117_2	GTTTCATATTTTTAATCC	GTttcatatttttaATCC	10
1099	1099_2	CCAAAAGAAACCAAACCC	CCaaaagaaaccaaACCC	88
1109	1109_2	TGAAACCATTACTACAACC	TGAaaccattactacaACC	22
1113	1113_2	CTATACCTAAAACAATCTA	CTatacctaaaacaaTCTA	86
1119	1119_2	CAAATATTCACAAATCCTA	CaaatattcacaaatCCTA	78
1125	1125_2	ACAATATATTCCTCAATCA	ACaatatattcctcaATCA	74
1127	1127_2	CATCCCTTTACCACTTT	CatccctttaccaCTTT	97
1118	1118_2	TAATATCCTCATTACCCATT	TaatatcctcattaccCATT	97
1103	1103_2	TCTATTCCTTAACCCAAC	TCtattccttaacccCAAC	81
1122	1122_2	AATCTTATTTACATCTTCC	AATCttatttacatcttCC	11
1126	1126_2	CCTGTAACAATTATACA	CCTGtaacaattataCA	93
1122	1122_3	AATCTTATTTACATCTTCC	AatcttatttacaTCtTCC	54
1122	1122_4	AATCTTATTTACATCTTCC	AAtcTtatttacAtCttCC	17
1122	1122_5	AATCTTATTTACATCTTCC	AAtcttatttacAtCttCC	21
1122	1122_6	AATCTTATTTACATCTTCC	AatctTatttacaTCttCC	12
1122	1122_7	AATCTTATTTACATCTTCC	AatcttatttacAtCttCC	28
1122	1122_8	AATCTTATTTACATCTTCC	AAtcttatttacAtcTTCC	28
1122	1122_9	AATCTTATTTACATCTTCC	AAtcTtatttacAtctTCC	11
1122	1122_10	AATCTTATTTACATCTTCC	AatctTatttacAtctTCC	9
1122	1122_11	AATCTTATTTACATCTTCC	AatcTtatttacatcTTCC	10
1122	1122_12	AATCTTATTTACATCTTCC	AATcTtatttacAtcTtCC	10
1122	1122_13	AATCTTATTTACATCTTCC	AatCTtatttacAtcttCC	10
1122	1122_14	AATCTTATTTACATCTTCC	AatCttatttacatctTCC	7
1122	1122_15	AATCTTATTTACATCTTCC	AatcttatttacaTCttCC	32
1122	1122_16	AATCTTATTTACATCTTCC	AatCttatttacatcTTCC	4
1122	1122_17	AATCTTATTTACATCTTCC	AAtCttatttacatcTtCC	5
1122	1122_18	AATCTTATTTACATCTTCC	AaTcTtatttacaTcTtCC	9

1122	1122_19	AATCTTATTTACATCTTCC	AatcTTatttacatcTtCC	5
1122	1122_20	AATCTTATTTACATCTTCC	AatcTtatttacatCttCC	13
1122	1122_21	AATCTTATTTACATCTTCC	AAtcttatttacatCttCC	23
1122	1122_22	AATCTTATTTACATCTTCC	AatctTatttacatCttCC	8
1122	1122_23	AATCTTATTTACATCTTCC	AatcTTatttacatCttCC	4
1122	1122_24	AATCTTATTTACATCTTCC	AatctTatttacatcTTCC	8
1122	1122_25	AATCTTATTTACATCTTCC	AATcTTatttacatcTtCC	5
1122	1122_26	AATCTTATTTACATCTTCC	AAtctTatttacatcTtCC	12
1122	1122_27	AATCTTATTTACATCTTCC	AaTCTtatttacatcTtCC	3
1122	1122_28	AATCTTATTTACATCTTCC	AaTcTTatttacatcTtCC	3
1122	1122_29	AATCTTATTTACATCTTCC	AatCTTatttacatcTtCC	3
1122	1122_30	AATCTTATTTACATCTTCC	AAtcTTatttacatctTCC	5
1122	1122_31	AATCTTATTTACATCTTCC	AAtcTtatttacatctTCC	12
1122	1122_32	AATCTTATTTACATCTTCC	AAtcttatttacatctTCC	33
1122	1122_33	AATCTTATTTACATCTTCC	AatCtTatttacatctTCC	3
1122	1122_34	AATCTTATTTACATCTTCC	AatcTTatttacatctTCC	6
1122	1122_35	AATCTTATTTACATCTTCC	AatcTtatttacatctTCC	16
1122	1122_36	AATCTTATTTACATCTTCC	AATCtTatttacatcttCC	8
1122	1122_37	AATCTTATTTACATCTTCC	AAtCTTatttacatcttCC	5
1122	1122_38	AATCTTATTTACATCTTCC	AAtCttatttacatcttCC	16
1122	1122_39	AATCTTATTTACATCTTCC	AaTCTtatttacatcttCC	7
1122	1122_40	AATCTTATTTACATCTTCC	AaTCtTatttacatcttCC	5
1122	1122_41	AATCTTATTTACATCTTCC	AatCTTatttacatcttCC	5
1122	1122_42	AATCTTATTTACATCTTCC	AatCTtatttacatcttCC	13
1122	1122_43	AATCTTATTTACATCTTCC	AatcTTatttacatcttCC	17
1109	1109_3	TGAAACCATTACTACAACC	TgaaaccattacTAcaaCC	58
1109	1109_4	TGAAACCATTACTACAACC	TgaaaccattacTAcAaCC	20
1109	1109_5	TGAAACCATTACTACAACC	TgaAAccattacTacAaCC	23
1109	1109_6	TGAAACCATTACTACAACC	TgAaAccattactAcaaCC	50
1109	1109_7	TGAAACCATTACTACAACC	TgAaaCcattactAcaaCC	46
1109	1109_8	TGAAACCATTACTACAACC	TgaAAccattacTacaaCC	48
1109	1109_9	TGAAACCATTACTACAACC	TgaaaccattactaCAaCC	25
1109	1109_10	TGAAACCATTACTACAACC	TgaaAccattacTaCaACC	24
1109	1109_11	TGAAACCATTACTACAACC	TGaaAccattactaCaaCC	36
1109	1109_12	TGAAACCATTACTACAACC	TgAAAccattactaCaaCC	20
1109	1109_13	TGAAACCATTACTACAACC	TgAAaCcattactaCaaCC	26
1109	1109_14	TGAAACCATTACTACAACC	TgAaaccattactaCaaCC	27
1109	1109_15	TGAAACCATTACTACAACC	TGaAaccattacTacAaCC	14
1109	1109_16	TGAAACCATTACTACAACC	TgAaaCcattactacAACC	12
1109	1109_17	TGAAACCATTACTACAACC	TgaaaCcattacTacAaCC	36
1109	1109_18	TGAAACCATTACTACAACC	TgaaaCcattacTacaaCC	62
1109	1109_19	TGAAACCATTACTACAACC	TGaaAccattactacaaCC	47
1109	1109_20	TGAAACCATTACTACAACC	TgaAaccattactaCAaCC	19
1109	1109_21	TGAAACCATTACTACAACC	TgaAaccattactACaACC	16

1109	1109_22	TGAAACCATTACTACAACC	TgAAaccattactACaACC	9
1109	1109_23	TGAAACCATTACTACAACC	TgAaAccattactAcaACC	29
1109	1109_24	TGAAACCATTACTACAACC	TgaaaCcattactAcaACC	41
1109	1109_25	TGAAACCATTACTACAACC	TgaAACcattactAcaaCC	34
1109	1109_26	TGAAACCATTACTACAACC	TgaAaCcattactaCaaCC	28
1109	1109_27	TGAAACCATTACTACAACC	TGaAaCcattactacAACC	10
1109	1109_28	TGAAACCATTACTACAACC	TgAAaCcattactAcAACC	52
1109	1109_29	TGAAACCATTACTACAACC	TGaAAccattactacaACC	16
1109	1109_30	TGAAACCATTACTACAACC	TGAaaccattactacaaCC	36
1109	1109_31	TGAAACCATTACTACAACC	TgaaaCcattactaCaACC	21
1109	1109_32	TGAAACCATTACTACAACC	TgAAAccattactacAACC	9
1109	1109_33	TGAAACCATTACTACAACC	TgAaaCcattactacAaCC	14
1109	1109_34	TGAAACCATTACTACAACC	TGaaaccattactacaACC	43
1109	1109_35	TGAAACCATTACTACAACC	TgAAaCcattactacaACC	15
1109	1109_36	TGAAACCATTACTACAACC	TgaAACcattactacaaCC	15
1109	1109_37	TGAAACCATTACTACAACC	TGaAaCcattactacaaCC	16
1109	1109_38	TGAAACCATTACTACAACC	TGaaaCcattactacaaCC	38
1109	1109_39	TGAAACCATTACTACAACC	TgAAACcattactacaaCC	14
1109	1109_40	TGAAACCATTACTACAACC	TgAAaCcattactacaaCC	16
1109	1109_41	TGAAACCATTACTACAACC	TgaAaCcattactacaaCC	28
1109	1109_42	TGAAACCATTACTACAACC	TgaaACcattactacaaCC	28
1122	1122_44	AATCTTATTTACATCTTCC	AatcttatttacaTCTtCC	65
1122	1122_45	AATCTTATTTACATCTTCC	AatcTtatttacAtCttCC	38
1122	1122_46	AATCTTATTTACATCTTCC	AatcTtatttacaTcTTCC	34
1122	1122_47	AATCTTATTTACATCTTCC	AAtCttatttacAtcTtCC	10
1122	1122_48	AATCTTATTTACATCTTCC	AAtcTtatttacATcTtCC	35
1122	1122_49	AATCTTATTTACATCTTCC	AatCttatttacAtcTtCC	10
1122	1122_50	AATCTTATTTACATCTTCC	AAtCttatttacAtcttCC	11
1122	1122_51	AATCTTATTTACATCTTCC	AAtctTatttacatCTtCC	9
1122	1122_52	AATCTTATTTACATCTTCC	AatcTTatttacAtcTtCC	12
1122	1122_53	AATCTTATTTACATCTTCC	AatctTatttacatCTtCC	8
1122	1122_54	AATCTTATTTACATCTTCC	AaTcTtatttacatcTTCC	4
1122	1122_55	AATCTTATTTACATCTTCC	AAtcttatttacAtcTtCC	27
1122	1122_56	AATCTTATTTACATCTTCC	AAtCtTatttacAtcttCC	5
1122	1122_57	AATCTTATTTACATCTTCC	AAtcTTatttacatcttCC	14
1122	1122_58	AATCTTATTTACATCTTCC	AaTCttatttacatcttCC	13
1122	1122_59	AATCTTATTTACATCTTCC	AATcttatttacatCttCC	6
1122	1122_60	AATCTTATTTACATCTTCC	AAtcTtatttacatCttCC	10
1122	1122_61	AATCTTATTTACATCTTCC	AAtcTTatttacatcTtCC	6
1122	1122_62	AATCTTATTTACATCTTCC	AatCtTatttacatcTtCC	3
1122	1122_63	AATCTTATTTACATCTTCC	AATCttatttacaTcttCC	5
1122	1122_64	AATCTTATTTACATCTTCC	AatCttatttacatcTtCC	7
1122	1122_65	AATCTTATTTACATCTTCC	AatCttatttacatcttCC	32
1122	1122_66	AATCTTATTTACATCTTCC	AatcttatttacatcTTCC	19

1122	1122_67	AATCTTATTTACATCTTCC	AATCttatttacatcTtCC	3
1122	1122_68	AATCTTATTTACATCTTCC	AATcTtatttacatcTtCC	4
1122	1122_69	AATCTTATTTACATCTTCC	AAtCTtatttacatcTtCC	3
1122	1122_70	AATCTTATTTACATCTTCC	AAtCtTatttacatcTtCC	3
1122	1122_71	AATCTTATTTACATCTTCC	AAtcTtatttacatcTtCC	13
1122	1122_72	AATCTTATTTACATCTTCC	AaTCttatttacatcTtCC	5
1122	1122_73	AATCTTATTTACATCTTCC	AatCTtatttacatcTtCC	5
1122	1122_74	AATCTTATTTACATCTTCC	AatctTatttacatcTtCC	10
1122	1122_75	AATCTTATTTACATCTTCC	AAtCTtatttacatctTCC	3
1122	1122_76	AATCTTATTTACATCTTCC	AAtCttatttacatctTCC	5
1122	1122_77	AATCTTATTTACATCTTCC	AaTCttatttacatctTCC	5
1122	1122_78	AATCTTATTTACATCTTCC	AatCTtatttacatctTCC	4
1122	1122_79	AATCTTATTTACATCTTCC	AAtCTtatttacatcttCC	7
1122	1122_80	AATCTTATTTACATCTTCC	AAtCtTatttacatcttCC	5
1122	1122_81	AATCTTATTTACATCTTCC	AatCtTatttacatcttCC	8
1109	1109_43	TGAAACCATTACTACAACC	TgAAaccattacTAcAaCC	18
1109	1109_44	TGAAACCATTACTACAACC	TgAaAccattacTacAaCC	27
1109	1109_45	TGAAACCATTACTACAACC	TgaAaCcattacTacAaCC	65
1109	1109_46	TGAAACCATTACTACAACC	TgAaaccattacTacaACC	25
1109	1109_47	TGAAACCATTACTACAACC	TgaAaccattacTacaACC	35
1109	1109_48	TGAAACCATTACTACAACC	TgaaAccattacTacaACC	48
1109	1109_49	TGAAACCATTACTACAACC	TgaAaCcattacTacaaCC	44
1109	1109_50	TGAAACCATTACTACAACC	TgaAaccattacTaCaaCC	34
1109	1109_51	TGAAACCATTACTACAACC	TGaaaccattacTacaACC	29
1109	1109_52	TGAAACCATTACTACAACC	TgAAaccattacTacaACC	23
1109	1109_53	TGAAACCATTACTACAACC	TgaaaCcattacTaCaaCC	39
1109	1109_54	TGAAACCATTACTACAACC	TGaaaccattactaCaaCC	33
1109	1109_55	TGAAACCATTACTACAACC	TgAaAccattactaCaaCC	29
1109	1109_56	TGAAACCATTACTACAACC	TGaaAccattactacAACC	16
1109	1109_57	TGAAACCATTACTACAACC	TGaaAccattactacAaCC	18
1109	1109_58	TGAAACCATTACTACAACC	TgAaACcattactacaaCC	12
1109	1109_59	TGAAACCATTACTACAACC	TgAaaccattactaCAaCC	13
1109	1109_60	TGAAACCATTACTACAACC	TgaaAccattactACaaCC	36
1109	1109_61	TGAAACCATTACTACAACC	TGaaaccattactAcaACC	34
1109	1109_62	TGAAACCATTACTACAACC	TgAaaCcattactACaaCC	43
1109	1109_63	TGAAACCATTACTACAACC	TGaAAccattactaCaaCC	19
1109	1109_64	TGAAACCATTACTACAACC	TGaaaCcattactACaaCC	29
1109	1109_65	TGAAACCATTACTACAACC	TGaAaccattactAcaaCC	40
1109	1109_66	TGAAACCATTACTACAACC	TgaAAccattactAcAACC	14
1109	1109_67	TGAAACCATTACTACAACC	TGaAaccattactAcAaCC	14
1109	1109_68	TGAAACCATTACTACAACC	TGaaaCcattactAcAaCC	27
1109	1109_69	TGAAACCATTACTACAACC	TgAaaCcattactAcAACC	31
1109	1109_70	TGAAACCATTACTACAACC	TgAaAccattactAcAaCC	24
1109	1109_71	TGAAACCATTACTACAACC	TgaaACcattactacAACC	10

1109	1109_72	TGAAACCATTACTACAACC	TGAaaccattactacAaCC	11
1109	1109_73	TGAAACCATTACTACAACC	TgaAACcattactAcAaCC	34
1109	1109_74	TGAAACCATTACTACAACC	TGaAaCcattactacaACC	15
1109	1109_75	TGAAACCATTACTACAACC	TGaaACcattactacaaCC	14
1109	1109_76	TGAAACCATTACTACAACC	TGaAaccattactaCaaCC	22
1109	1109_77	TGAAACCATTACTACAACC	TgaAAccattactaCaaCC	30
1109	1109_78	TGAAACCATTACTACAACC	TgaaAccattactaCaaCC	50
1109	1109_79	TGAAACCATTACTACAACC	TgaAACcattactacAaCC	9
1109	1109_80	TGAAACCATTACTACAACC	TGaAaccattactacaaCC	31
1109	1109_81	TGAAACCATTACTACAACC	TgAaaCcattactacaaCC	31

In the oligonucleotide compound column, capital letters represent beta-D-oxy LNA nucleosides, LNA cytosines are 5-methyl cytosine, lower case letters are DNA nucleosides, and all internucleoside linkages are phosphorothioate.

Example 5 Testing in vitro efficacy of LNA oligonucleotides in iCell GlutaNeurons at 25μM

An oligonucleotide screen was performed in a human cell line using selected LNA oligonucleotides from the previous examples.

The iCell GlutaNeurons derived from human induced pluripotent stem cell were purchased from the vendor listed in table 3, and were maintained as recommended by the supplier in a humidified incubator at 37°C with 5% CO_2. For the screening assays, cells were seeded in 96 multi well plates in media recommended by the supplier (see table 3 in the Materials and Methods section). The number of cells/well was optimized (Table 3).

Cells were grown for 7 days before addition of the oligonucleotide in concentration of 25 μM (dissolved in medium). 4 days after addition of the oligonucleotide, the cells were harvested. RNA extraction and qPCR was performed as described for “Example 1”

Primer assays for ATXN3 and house keeping gene were:

ATXN3 primer assay (Assay ID: N/A, Item Name: Hs.PT.58.39355049):

Forward primer: GTTTCTAAAGACATGGTCACAGC (SEQ ID NO 1128)

Reverse: CTATCAGGACAGAGTTCACATCC (SEQ ID NO 1129)

Probe: 56-FAM/AAAGGCCAG/ZEN/CCACCAGTTCAGG/3IABkFQ/ (SEQ ID NO 1030)

TBP primer assay (Assay ID: N/A, Item name: Hs.PT.58v.39858774

Probe: 5’- /5HEX/TGA TCT TTG /ZEN/CAG TGA CCC AGC ATC A/3IABkFQ/ -3’ (SEQ ID NO 1131)

Primer 1: 5’- GCT GTT TAA CTT CGC TTC CG-3’ (SEQ ID NO 1132)

Primer 2: 5’- CAG CAA CTT CCT CAA TTC CTT G-3’ (SEQ ID NO 1133)

As part of the screening cascade 1170 compounds were evaluated in the cell lines SK-N-AS and A431 where compound efficacy was evaluated (Tables 4 - 6). Of these, 50 of the most effective compounds were evaluated for caspase activation of which 18 underwent further evaluation in the descrived in the three other in vitro tox assays (cumulative score is shown in Table 7).

Conclusively, 8 compounds were identified as being highly effective and potent in vitro, and with a low or absent toxicity in the 4 in vitro assays – these compounds were therefore selected for evaluated in transgenic mice expressing human ATNX3 pre-mRNA: Compounds # 1856_1, 1813_1, 1812_1, 1809_2, 1607_1, 1122_62, 1122_67 and 1122_33.

Table 7 – Data obtained from examples 5, 6 & 7

CMPID	Total tox score	SK-N-AS EC50 (µM)	A-431 EC50 (µM)	HiPSC EC50 (µM)	HiPCS, Maximal efficacy at 25µM (% remaining ATXN3 transcript)
1856_1	1,5	0,53	0,22	0,23	2,87
1806_2	2	0,35	0,19	0,03	0,91
1888_1	-	0,72	0,54	-	
1813_1	2	0,24	0,08	0,04	1,85
1640_1	-	1,50	0,19	-	
1812_1	1,5	0,20	0,09	0,09	0,59
1117_2	-	0,73	0,57	-	
1810_1	-	0,36	0,14	-	
1809_2	1,25	0,22	0,09	0,05	1,44
1489_1	-	1,16	0,30	-	
1867_1	-	0,54	0,50	-	
1893_1	-	0,95	0,34	0,41	4
1906_1	-	0,36	0,57	0,04	2,55
1214_1	-	1,05	0,38	-	
1213_1	-	1,01	0,38	-	
1423_1	-	0,75	0,23	0,03	3,58
1790_1	-	0,42	0,47	-	
1605_1	-	0,47	0,17	-	
1607_1	2,5	0,32	0,25	0,08	4,46
1805_1	-	0,75	0,23	-	
1806_1	-	0,45	0,20	0,04	1,3
1809_1	3	0,24	0,20	0,02	1,81
1808_1	2	0,26	0,22	0,06	1,4
1625_1	0,5	0,94	0,25	0,66	7,16
1122_54	-	0,62	0,15	-	
1122_16	-	0,30	0,15	-	
1122_17	-	0,33	0,17	0,11	1,07
1122_62	0,5	0,21	0,10	0,03	3,53
1122_19	-	0,28	0,24	-	

1122_23	-	0,54	0,18	0,05	0,59
1122_67	0	0,29	0,10	0,01	0,52
1122_68	-	0,28	0,13	0,01	
1122_69	-	0,27	0,12	-	
1122_70	-	0,20	0,10	-	
1122_27	1	0,23	0,12	0,03	0,55
1122_72	0,5	0,25	0,15	0,06	2,28
1122_28	1	0,20	0,12	0,01	0,37
1122_29	-	0,19	0,09	0,02	1,6
1122_73	-	0,29	0,18	0,04	1,59
1122_75	1	0,44	0,12	0,03	2
1122_76	-	0,33	0,19	-	
1122_77	1	0,30	0,20	0,04	1,97
1122_78	-	0,29	0,18	0,02	1,91
1122_33	1,25	0,18	0,10	0,02	1,84
1122_37	-	0,25	0,13	0,03	0,89
1122_80	-	0,33	0,17	-	
1122_41	-	0,24	0,16	0,01	0,47
1109_22	-	0,90	0,23	0,11	8,41
1109_32	0	0,75	0,17	0,09	3,49
1109_79	-	1,48	0,20	-	

Example 7: *In vivo* transgenic mouse Study

Animal Care

In vivo activity and tolerability of the compounds were tested in 10 - 13 week old B6;CBA-Tg(ATXN3*)84.2Cce/IbezJ male and female mice (JAX® Mice,The Jackson Laboratory) housed 3-5 per cage. The mice are transgenic mice which express the human ATXN3 pre-mRNA sequence, with 84 CAG repeats motif, an allele which is associated with MJD in humans). Animals were held in colony rooms maintained at constant temperature (22 ± 2°C) and humidity (40 + 80%) and illuminated for 12 hours per day (lights on at 0600 hours). All animals had ad libitum access to food and water throughout the studies. All procedures are performed in accordance with the respective Swiss regulations and approved by the Cantonal Ethical Committee for Animal Research.

Administration Route -Intra-cisterna Magna injections.

The compounds were administered to mice by intra cisterna magna (ICM) injections. Prior to ICM injection the animals received 0.05 mg/kg Buprenorphine dosed sc as analgesia. For the ICM injection animals were placed in isofluran. Intracerebroventricular injections were performed using a Hamilton micro syringe with a FEP catheter fitted with a 36 gauge needle. The skin was incised, muscles retracted and the atlanto-occipital membrane exposed. Intracerebroventricular injections were performed using a Hamilton micro syringe with a catheter fitted with a 36 gauge needle. The 4 microliter bolus of test compound or vehicle was injected over 30 seconds. Muscles were repositioned and skin closed with 2-3 sutures. Animals were placed in a varm environment until they recovered from the procedure.

2 independent experiments were performed with groups of different compounds as shown in Table 8.

Table 8

Compound ID	Dose, µg	Time-point	Group Size
Saline only	0	4wk	6
1856_1	250	4wk	8
1813_1	250	4wk	8
1812_1	250	4wk	8
1809_2	250	4wk	8
1607_1	250	4wk	8
1122_62	250	4wk	8
1122_67	250	4wk	8
1122_33	250	4wk	8

Tolerability Results:

All compounds were found to be tolerated up to the 4 weeks timepoint. Acute toxicity was measured by monitoring the animal's behavior as described in WO2016/126995 (see example 9). Sub-acute toxicity was measured by monitoring the body weight of each animal during the time course of the experiment, with >5% weight reduction indicative of sub-acute toxicity. In some groups 1 or 2 animals did show some distress after the ICM administration and were euthanized, but this was likely to be due to the procedure rather than a adverse toxicity of any of the compounds. All eight compounds were therefore considered to be well tolerated in vivo.

4 weeks post administration, the animals were sacrificed, and tissues from the cortex, midbrain, cerebellum, hippocampus pons/medulla and striatum were collected weighed and snap frozen in liquid N2 directly after sampling. Samples were stored on dry ice until storage at -80 °C.

Analysis of *in vivo* samples. Description of tissue preparation for content measurement and qPCR.

Table 9

Compound	EC50 in hiPSC-derived neurons, nM				
	Day 4	Day 6	Day 9	Day 12	Day 20
1122_67	7,2	1,3	1,4	1,1	1,1
1813_1	23	6,3	10	8,9	7,7
1100673	110	27	30	34	44
1101657	515	204	69	90	73
1102130	315	164	390	101	133
1103014	662	64	435	98	369
1102987	944	305	135	391	200

Compounds 1122_67 and 1813_1 were remarkably more potent than the 5 reference compounds, with compound 1122_67 being the most potent compound at all time points and both 1122_67 and 1813_1 gave a remarkably effective and long lasting inhibition of ATXN3 mRNA.

Example 9: Comparative *In vivo* transgenic mouse study

A further *in vivo* study was performed at Charles River Laboratories Den Bosch B.V., Groningen, NL, using compound 1122_67 and 1813_1, and reference compound 1100673 (WO2019/217708). The study used male and female B6;CBA-Tg(ATXN3*)84.2Cce/IbezJ mice with the compounds administered via intracisternal (ICM) administration. At two timepoints after compound administration, 1 or 4 weeks, animals were euthanized and terminal plasma samples and tissues were collected.

Animal Care

In vivo activity and tolerability of the compounds were tested in 62 B6;CBA-Tg(ATXN3*)84.2Cce/IbezJ male and female mice (JAX® Mice,The Jackson Laboratory) at the age between 7-10 weeks. Following arrival, animals were housed in groups up to 5 in individually vented cages (IVC, 40 x 20 x 16 cm) in a temperature (22 ± 2 °C) and humidity (55 ± 15%) controlled environment on a 12 hour light cycle (07.00 – 19.00h). Males and females were kept in separate cages. Standard diet (SDS Diets, RM1 PL) and domestic quality mains water were available ad libitum. If required, animals received soaked chow and/or Royal Canin in addition to Standard diet as part of pamper care. The experiments were conducted in strict accordance with the Guide for the Care and Use of Laboratory Animals (National Research Council 2011) and were in accordance with European Union directive 2010/63 and

Table 8

	Cortex		Midbrain		Cerebellum		Hippocampus		Pons/medulla		Striatum	
Compounds	EC50 (nM)	Max efficacy (% remaining)	EC50 (nM)	Max efficacy (% remaining)	EC50 (nM)	Max efficacy (% remaining)	EC50 (nM)	Max efficacy (% remaining)	EC50 (nM)	Max efficacy (% remaining)	EC50 (nM)	Max efficacy (% remaining)
1856_1	251	33	77	20	434	49	202	41	-	24	103	27
1813_1	260	22	93	20	347	47	279	30	-	22	89	18
1812_1	307	52	156	28	603	50	233	35	-	26	184	32
1809_2	134	57	153	34	511	50	111	46	-	21	93	29
1607_1	193	40	89	17	120	42	81	21	-	15	63	26
1122_62	125	56	74	26	226	16	86	46	-	19	54	36
1122_67	125	23	79	14	261	27	146	22	-	13	88	19
1122_33	102	47	38	16	166	35	79	24	-	17	63	29

All compounds tested gave efficacious target inhibition in the tissues tested and were tolerated at the doses tested. Compound 1122_33 across the compounds tested has either the best or second ranked highest specific activity (lower EC50) in all tissues, followed by 1122_62 and 1122_67.

Compounds 1122_67, 1607_1, 1813_1 and 1122_33 provided high efficacy *in vivo* in all tissues tested, illustrating a remarkable consistent inhibition of ATXN3 expression across the brain tissues tested. Based on an accumulative rank score compound 1122_67 was consistently either the best or second ranked compound in terms of efficacy of ATXN3 knock down in the tissues tested.

Table 10

Tissue		Cortex (A1)		Cerebellum		Brainstem		Midbrain		Striatum	
		EC50 (nM)	Max KD observed	EC50 (nM)	Max KD observed	EC50 (nM)	Max KD observed	EC50 (nM)	Max KD observed	EC50 (nM)	Max KD observed
1 week of treatment	1122_67	242	88%	833	74%	196	87%	165	89%	148	77%
	1813_1	278	61%	966	57%	377	85%	183	90%	118	51%
	1100673	391	67%	2012	48%	769	79%	279	81%	331	69%
4 week of treatment	1122_67	100	92%	365	81%	81	93%	94	95%	46	89%
	1813_1	ND	ND	ND	ND	ND	ND	ND	ND	ND	ND
	1100673	199	49%	1229	33%	419	72%	129	74%	130	35%

Tissue		Hippocampus		Spinal cord, cervical		Spinal cord, thoracic		Spinal cord, lumbar	
		EC50 (nM)	Max KD observed	EC50 (nM)	Max KD observed	EC50 (nM)	Max KD observed	EC50 (nM)	Max KD observed
1 week of treatment	1122_67	243	75%	41	89%	39	90%	54	89%
	1813_1	341	63%	45	90%	36	92%	48	91%
	1100673	516	66%	83	83%	51	83%	68	82%
4 week of treatment	1122_67	89	92%	16	93%	Imprecise	93%	18	93%
	1813_1	ND	ND	ND	ND	ND	ND	ND	ND
	1100673	329	52%	48	83%	Imprecise	84%	56	84%

FIGURE 9

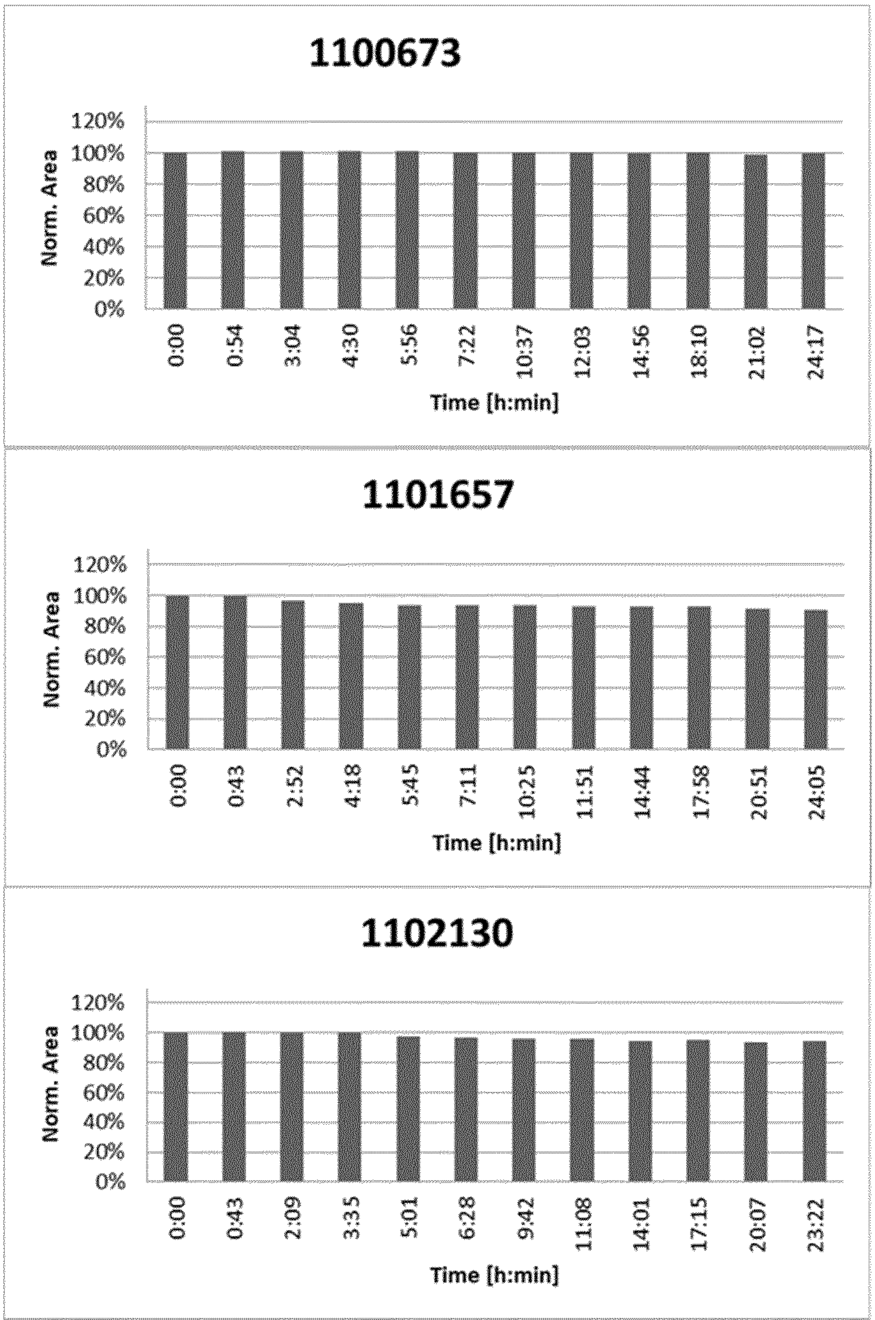
1100673
Norm. Area
120%
100%
80%
60%
40%
20%
0%
0:00
0:54
3:04
4:30
5:56
7:22
10:37
12:03
14:56
18:10
21:02
24:17
Time [h:min]
1101657
Norm. Area
120%
100%
80%
60%
40%
20%
0%
0:00
0:43
2:52
4:18
5:45
7:11
10:25
11:51
14:44
17:58
20:51
24:05
Time [h:min]
1102130
Norm. Area
120%
100%
80%
60%
40%
20%
0%
0:00
0:43
2:09
3:35
5:01
6:28
9:42
11:08
14:01
17:15
20:07
23:22
Time [h:min]

FIGURE 9 (continued)

INTERNATIONAL SEARCH REPORT

International application No
PCT/EP2020/065401

A. CLASSIFICATION OF SUBJECT MATTER
INV. C12N15/113 A61K31/712 A61K31/7125 A61K31/7115
ADD.

According to International Patent Classification (IPC) or to both national classification and IPC

B. FIELDS SEARCHED

Minimum documentation searched (classification system followed by classification symbols)
A61K C12N

Documentation searched other than minimum documentation to the extent that such documents are included in the fields searched

Electronic data base consulted during the international search (name of data base and, where practicable, search terms used)

EPO-Internal, WPI Data, BIOSIS, COMPENDEX, EMBASE, FSTA

C. DOCUMENTS CONSIDERED TO BE RELEVANT

Category*	Citation of document, with indication, where appropriate, of the relevant passages	Relevant to claim No.
X	LAUREN R. MOORE ET AL: "Evaluation of Antisense Oligonucleotides Targeting ATXN3 in SCA3 Mouse Models", MOLECULAR THERAPY-NUCLEIC ACIDS, vol. 7, 1 June 2017 (2017-06-01), pages 200-210, XP055674189, US ISSN: 2162-2531, DOI: 10.1016/j.omtn.2017.04.005 figure 1 -----	1,10-17

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[X] Further documents are listed in the continuation of Box C.

[X] See patent family annex.

* Special categories of cited documents :

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"X" document of particular relevance; the claimed invention cannot be considered novel or cannot be considered to involve an inventive step when the document is taken alone

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"&" document member of the same patent family

Date of the actual completion of the international search	Date of mailing of the international search report
20 August 2020	31/08/2020

Name and mailing address of the ISA/	Authorized officer
European Patent Office, P.B. 5818 Patentlaan 2 NL - 2280 HV Rijswijk Tel. (+31-70) 340-2040, Fax: (+31-70) 340-3016	Piret, Bernard

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INTERNATIONAL SEARCH REPORT

International application No
PCT/EP2020/065401

C(Continuation). DOCUMENTS CONSIDERED TO BE RELEVANT

Category*	Citation of document, with indication, where appropriate, of the relevant passages	Relevant to claim No.
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International application No
PCT/EP2020/065401

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